

CHAPTER

8

COMMON PITFALLS IN EVERYDAY VIBRATION MEASUREMENT

8.0 INTRODUCTION

Good quality data is essential to every vibration program. Without proper representation of the actual vibration, the diagnostic results, conclusions and recommendations become meaningless. The following pointers are given to help avoid pitfalls which can plague a vibration PM program, sometimes rendering it almost ineffective.

8.1 GENERAL CONSIDERATION FOR OBTAINING CONSISTENT QUALITY DATA

The following observations are provided based on many years of vibration data collection field experience. Also, they are broken into categories which reflect the order in which PM programs or vibration data collection would be conducted.

8.11 CHOOSING MEASUREMENT LOCATIONS

- 1) Choose proper measurement locations. Figure 1 shows both good and poor measurement locations. Locations should be as close to the machine bearings as possible. Try to locate these points within the load zone of the bearing, particularly on sleeve bearing supported machines, as well as heavily loaded or low-speed rolling element bearing machines. Studies have shown that placing the pickup within the load zone of low-speed, heavy rolls can increase the measured amplitude by 100% or more.
- 2) Do not compromise safety for a vibration measurement. Install a permanently mounted transducer if necessary to prevent potential injury.
- 3) Take horizontal measurements as near to the horizontal shaft centerline as possible (parallel to the earth). Also, see that the transducer is mounted as close to the bearing as possible, particularly if high frequency banded acceleration measurements (such as spike energy) are to be taken. Here again, please see Figure 1.
- 4) Take vertical measurements in as close to the vertical shaft centerline as possible (plumb with respect to the earth). Also, see that the transducer is mounted as close to the bearing as possible.
- 5) Take axial measurements parallel to the shaft being supported by the bearing and make the measurement at the same location each time (see Figure 1). Do not take axial data at the 12:00 position on one survey; then at 3:00; 6:00; or 9:00 on successive surveys. Again position the transducer as close to the bearing as possible.
- 6) On pumps, do not mistake seal locations for bearing locations.

- 7) Do not take bearing measurements on a foundation or fabricated bases unless that is truly what you want.
- 8) Do not assign measurement locations to thin/weak sheet metal such as on some motor end bells. It is preferable in this case to take measurements on a substantial part of the motor such as the motor case (see Figure 1).

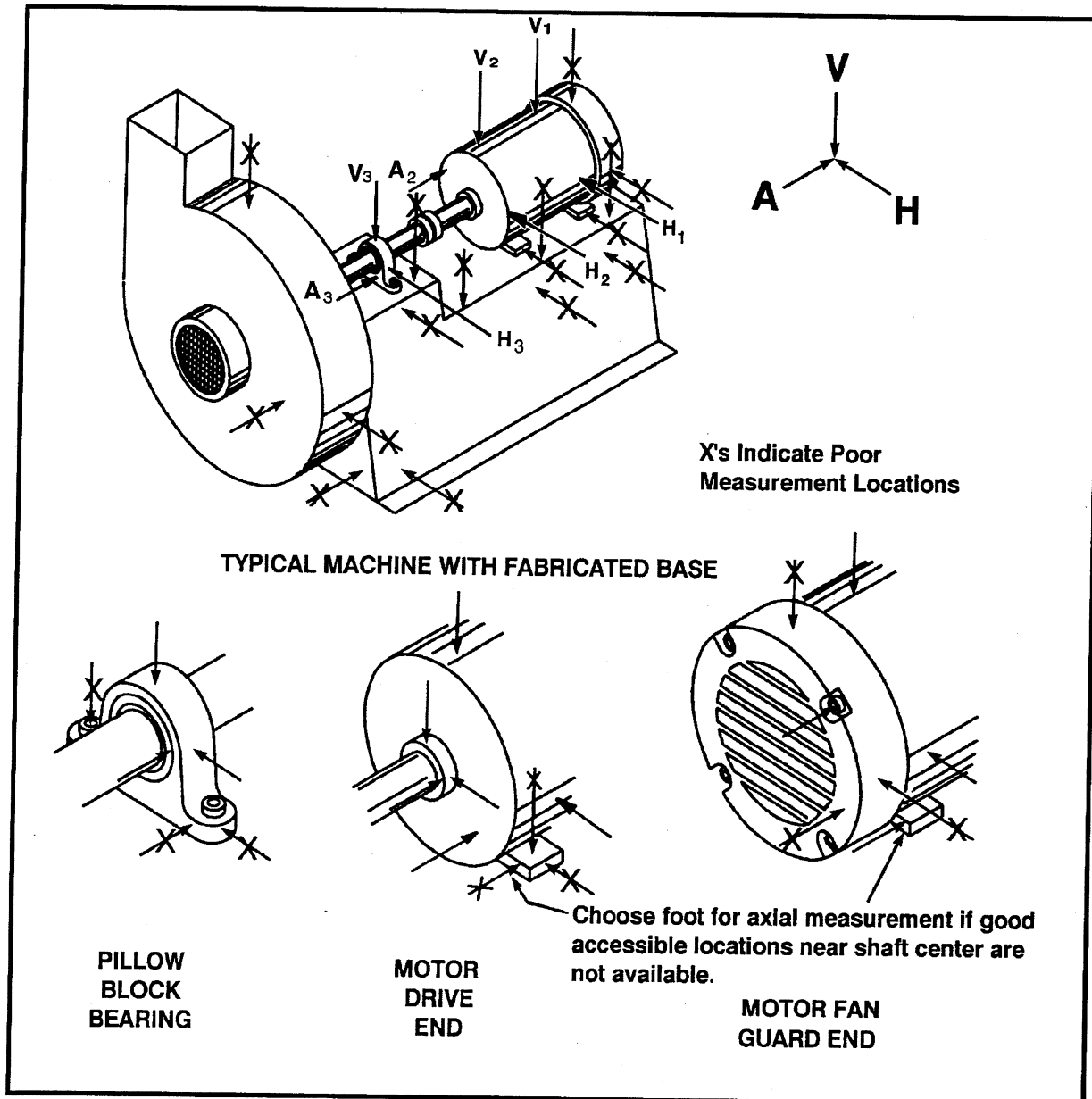


DIAGRAM SHOWING GOOD AND POOR MEASUREMENT LOCATIONS
FIGURE 1

8.12 MACHINE AND POINT IDENTIFICATION

1) Adopt naming and position identification conventions for all machines and points to be included in the PM program. Remain consistent as one moves from one plant area to another with these naming conventions. Table 1 shows one such naming convention. Please keep the following items in mind when adopting naming conventions and physically marking them on the machinery:

A) Number the measurement locations beginning from the physical outboard motor bearing to the outboard driven machine bearing (as shown in Figure 2). For example, assume a motor-driven fan. It is recommended that one of the following conventions be adopted as shown in Table 1 (or something similar):

TABLE I.
RECOMMENDED OPTIONAL MEASUREMENT LOCATION NAMING CONVENTIONS

LOCATION ON THE MACHINERY	NAMING CONVENTIONS	
	OPTION 1	OPTION 2*
MOTOR OUTBOARD BEARING (FAN SIDE)	1	MOV
MOTOR INBOARD BEARING (DRIVE SIDE)	2	MIV
FAN (DRIVEN) INBOARD BEARING	3	FIV
FAN (DRIVEN) OUTBOARD BEARING	4	FOV

*NOTE: OPTION 2 ASSUMES VELOCITY MEASUREMENTS. IF ACCELERATION IS MEASURED INSTEAD, AN "A" WOULD BE ENTERED AS THE THIRD LETTER.

Insert additional points in between these for equipment such as multi-stage gearboxes as needed.

B) Use consistent lettering conventions for the direction of the pickup (transducer) such as the following:

NAME MEASURING DIRECTION

A	Axial
H	Horizontal
V	Vertical

C) If measuring on a vertically oriented machine, such as a vertical pump, both of the radial measurements at each bearing will be in the horizontal direction. In this case it is necessary to know the north/south and east/west directions at the machine. If your PMP software will allow, input the point names as 1N, 1E (for bearing 1 in the north and east directions). Point 1A will be parallel to the shaft (or perpendicular to the earth). However, some PMP software only allows a point to be named with A, H, and V. Therefore, adopt a naming convention such as H being the east direction and V being the north direction. Usually, there is another input field in the PMP software for entering a point description. Use this field to tell the user which direction to measure. For instance, for point 1H, input "1 East" in the point description field and set the data collector to display this field when collecting data on a route.

2) Clearly mark the machine names with the name seen in the PMP computer database and data collector itself. This may require the use of a special paint which will withstand the environment in which the machine is located. On the other hand, if the machines themselves are periodically painted, or if the environment is too harsh, permanent metal stamps with the machine name imbedded on them may have to be used.

3) Clearly mark and label measurement point locations on each machine. Here again, metal stamps with the point identification may have to be used. Also, a washer may be permanently mounted (welded or glued) to the machine. This will not only provide a clearly located measurement point for consistent vibration data collection but also provide a good flat surface to mount the transducer or magnetic mount without worrying about it rocking on a rounded machine surface during data collection thereby rendering the data useless.

4) Provide machine drawings for each PMP machine showing measurement locations. See the sample machine drawing in Figure 2. Include important machine configuration information on the sketch such as bearing type and model number, gear tooth count, operating speed ranges of each component (RPM), blade or vane count on pumps/blowers, etc.

5) Construct a map of PMP machine locations within the plant. Figure 3 is a sample machine location map. This will help the person collecting data to verify that he is at the correct machine when collecting data as well as helping him set up an efficient route. This is crucial when many similar pieces of machinery are present in different locations in the plant such as air handlers or pumps. Typically, they are given the same name differentiated only by an equipment number. Make sure that the number is on the machine, in the PMP database, and on the machine location map.



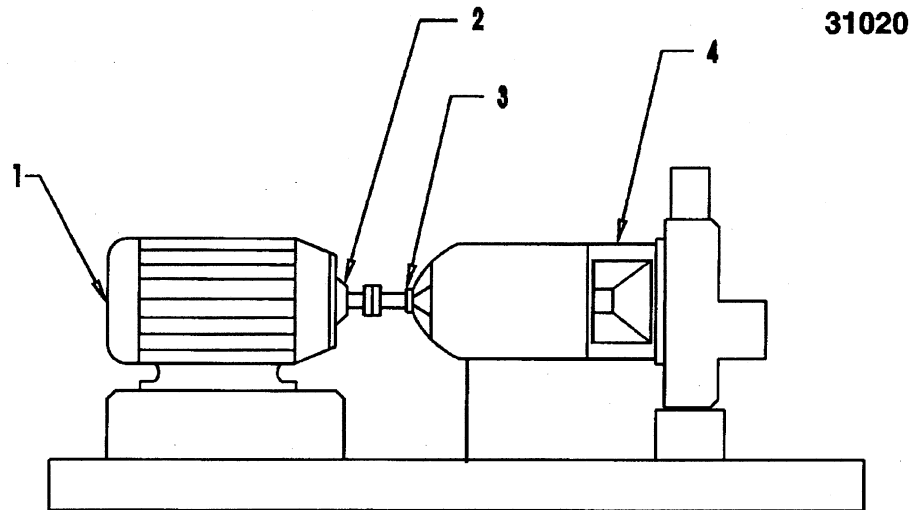
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MACHINERY DATA GUIDE SHEET

RELIABILITY SURVEY

predictive maintenance program

(PAGE__OF__)



MACHINE IDENT.: 3 COND XFER PUMP AREA: NGS-3
ROUTE: _____

DRIVER	INTERMEDIATE	DRIVEN
RPM min: 3560	RPM in:	RPM min: 3560
max:	out:	max:
MFGR: WESTINGHOUSE	RATIO:	MFGR: WESTINGHOUSE/PACIFIC
MODEL: 7209	MODEL:	MODEL:
SER NO.:	SER NO.:	SER NO.:
V 480 P 3 A 109	RATING:	RATING:
SHEAVE: DIA.	BELT# :	# BLADES:
BRG# : BALL 6312 (I)	BRG# : (I)	BRG# : BALL (I)
BRG# : BALL 6312 (O)	BRG# : (O)	BRG# : BALL (O)
H.P.: 100	GEAR T: CT.	GEAR T: CT.
OTHER:	OTHER:	OTHER:

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FIGURE 2
SAMPLE MACHINE DRAWING

8.13 MEASUREMENT PARAMETERS

1) Specify the optimum frequency span for each measurement point on each PMP machine. Guidelines recommending such frequency ranges will be given in Chapter 7. It is very important to know the vibration source frequencies such as bearing, gear mesh, blade pass, electrical line, and rotor bar pass frequencies that may occur with each machine. Also, harmonics of these frequencies must be considered. In general, if there is no gearbox involved, rolling element bearing machines would require F_{MAX} at about 40X to 50X RPM, while sleeve bearing machines would require a maximum frequency of only about 20X RPM (assuming speeds are above about 1500 RPM).

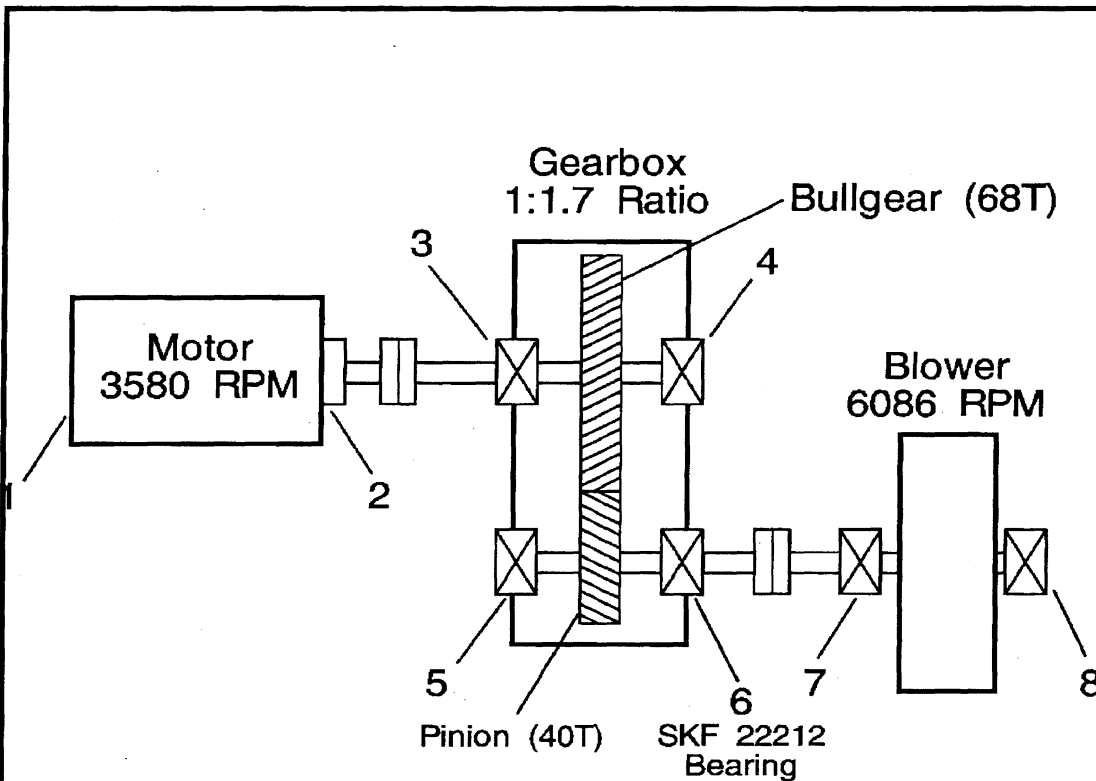
However, there will be special cases requiring much higher frequency spans. For example, consider the gear-driven blower shown in Figure 4. This blower is driven by a motor running at 3580 RPM. However, a gear unit is used to increase the blower shaft speed to 6086 RPM. Because one gear has 68 teeth and the other has 40 teeth, the gear mesh frequency (# teeth x RPM) is 243,440 CPM. Also, an SKF 22212 bearing is used on the output gear shaft which has bearing fault frequencies as noted in Figure 4.

Figure 5 shows a spectrum collected on the PMP route with F_{MAX} set to 12,000 CPM. This frequency span would encompass as many as 3 harmonics of motor speed, but only the operating speed frequency of the output gear and blower (6086 RPM). Note that nothing unusual is present in this spectrum. The motor, blower, and 2X motor frequencies are present, but not at levels which would typically cause concern and this machine would be reported to be in "Good" or "Fair" condition. However, this is not the case as will be shown.

Figure 6 shows the same measurement point (position 6 of Figure 4) on the same machine as the plot in Figure 5, however, F_{MAX} has been increased to 24,000 CPM. It shows that harmonics of the gearbox output frequency are present which indicate possible machine faults (such as slight looseness, or possibly a bearing turning on a shaft). Also, it is important to point out that with the higher F_{MAX} setting, the resolution (ability to differentiate peaks) has decreased. Therefore, more than one measurement may need to be taken at this point - one with a low F_{MAX} to differentiate 2X motor frequency (7160 CPM) from 2X electrical line frequency (7200 CPM) and another with a higher F_{MAX} to pick up the gear and bearing fault frequencies.

Figure 7 shows another plot from gearbox position 6 as in Figures 5 and 6. However, F_{MAX} has been significantly increased to 240,000 CPM (up to 39.4X blower RPM). Note now that 5 harmonics of the gearbox bearing outer race frequency (BPFO) have appeared, along with the bearing fundamental train frequency (FTF) sidebands, none of which were even present in the previous spectra (note that FTF is often referred to as the "cage frequency of a bearing"). This signifies an extremely dangerous situation which should be corrected as soon as possible. This spectra shows the SKF 22212 gearbox bearing approaching catastrophic failure, but it would not have even been noticed had F_{MAX} not been noticeably increased.

Figure 8 shows the same measurement point on the same machine as Figures 5 thru 7. However, F_{MAX} has been increased to 360,000 CPM. Now, the fundamental gear mesh frequency (GMF) has appeared for the first time as we went to higher and higher frequencies. Also, note the output gear shaft speed sidebands spaced on either side of GMF. It is quite common to see 1X gear mesh frequency in a spectrum, and is not necessarily considered a problem. The bearing defect is still clearly the most dominant problem noted by this spectrum.



SKF 22212 BEARING DEFECT FREQUENCY MULTIPLES (CPM) AT 6086 RPM

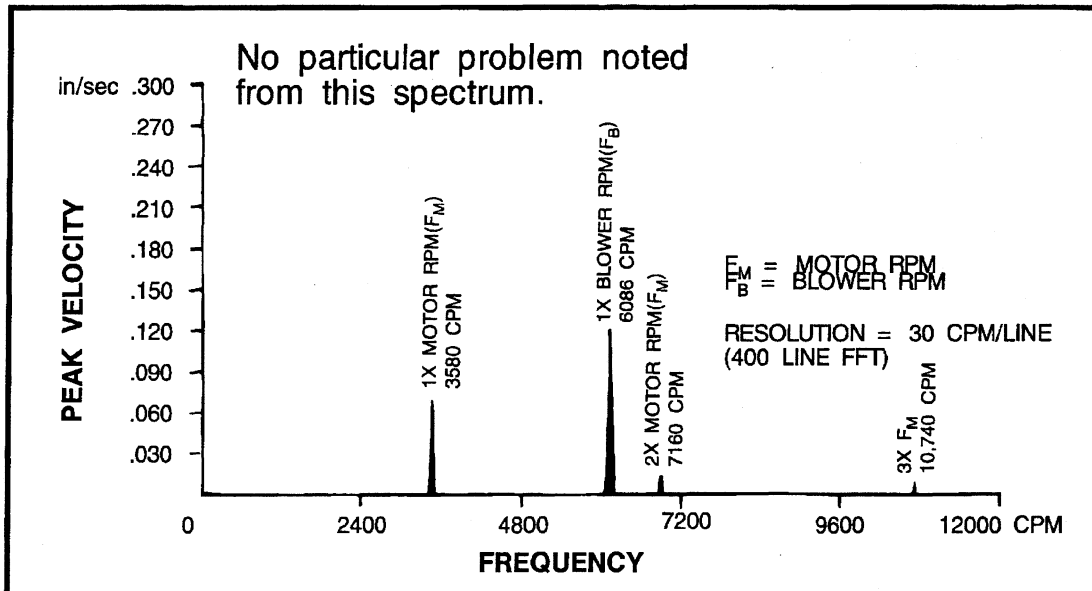
DEFECT FREQ.	DEFECT COEF.	BEARING DEFECT FREQUENCY MULTIPLES				
		1X	2X	3X	4X	5X
BPFO	7.26X	44,184	88,368	132,552	176,736	220,920
BPFI	9.72X	59,156	118,312	177,468	236,624	295,780
FTF	.43X	2,617	5,234	7,851	10,468	13,085
BSF	3.33X	20,266	40,532	60,798	81,064	101,330

GEAR MESH FREQUENCY MULTIPLES

1X GMF	243,440 CPM
2X GMF	486,880 CPM
3X GMF	730,320 CPM

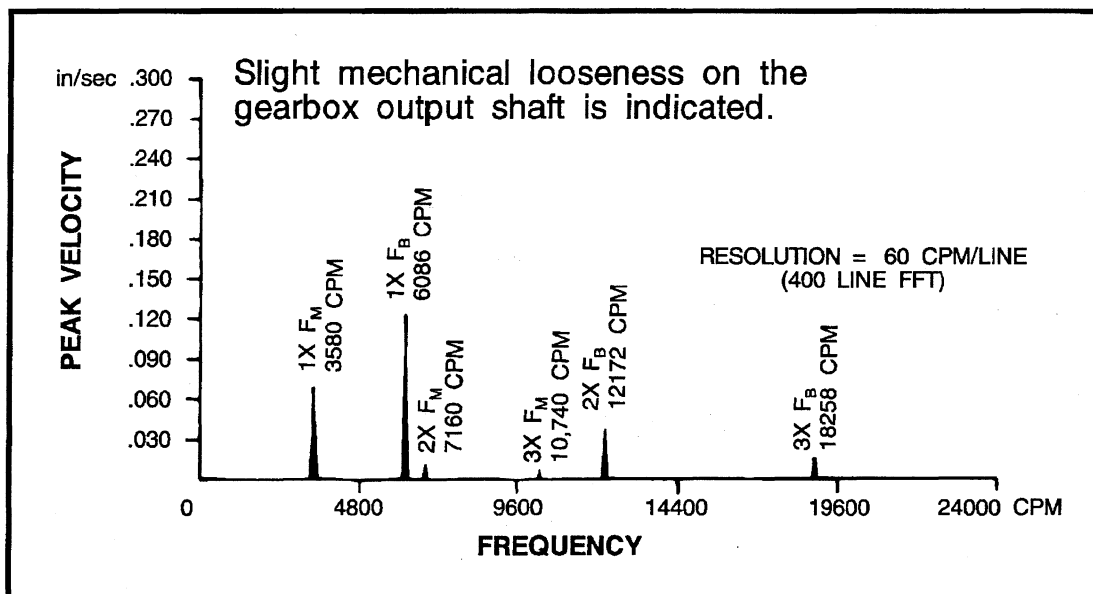
$$\begin{aligned}
 \text{GMF} &= \# \text{ TEETH} \times \text{RPM} \\
 &= 68\text{T} \times 3580 = 40\text{T} \times 6086 \\
 \text{GMF} &= \underline{\underline{243,440 \text{ CPM}}}
 \end{aligned}$$

MOTOR-DRIVEN GEARBOX AND BLOWER MACHINE DRAWING
FIGURE 4



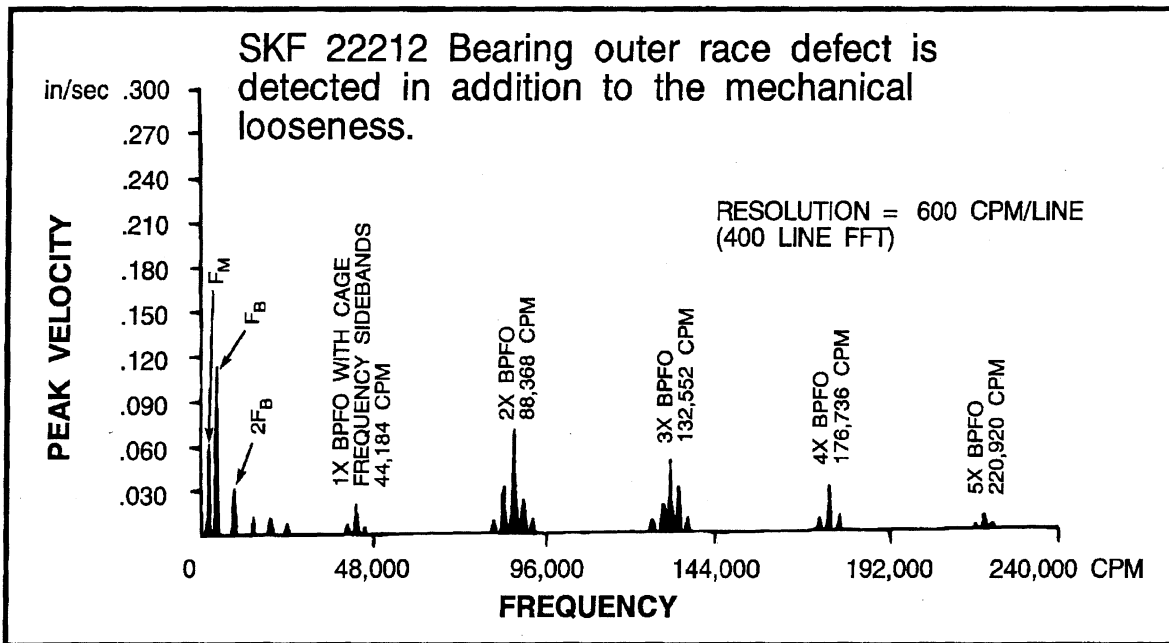
0-12,000 CPM SPECTRUM CAPTURES 1X THRU 3X GEARBOX INPUT SHAFT RPM, BUT ONLY 1X GEARBOX OUTPUT SHAFT RPM GOOD RESOLUTION OF 30 CPM/FFT LINE.

FIGURE 5



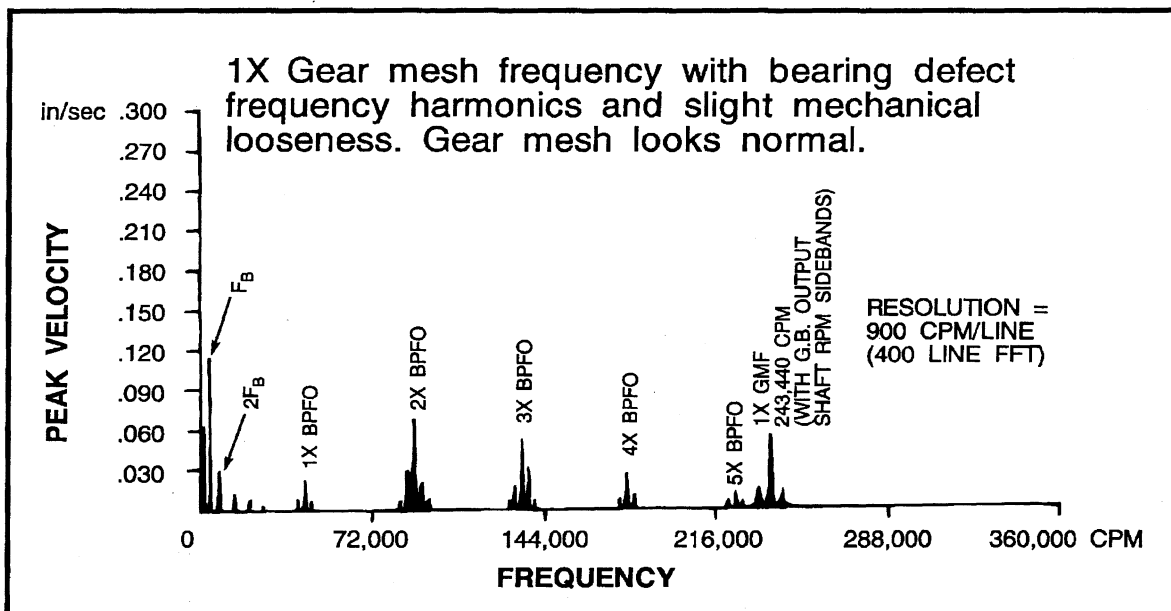
0-24,000 CPM SPECTRUM CAPTURES 3 HARMONICS OF GEARBOX OUTPUT SHAFT RPM IN ADDITION TO INFORMATION FROM FIGURE 5 ABOVE. RESOLUTION DECLINES SLIGHTLY.

FIGURE 6



0-240,000 CPM SPECTRUM CAPTURES 5 PINION BEARING OUTER RACE DEFECT FREQUENCY HARMONICS IN ADDITION TO INFORMATION FROM FIGURE 6.
RESOLUTION DECLINES TO 600 CPM/LINE.

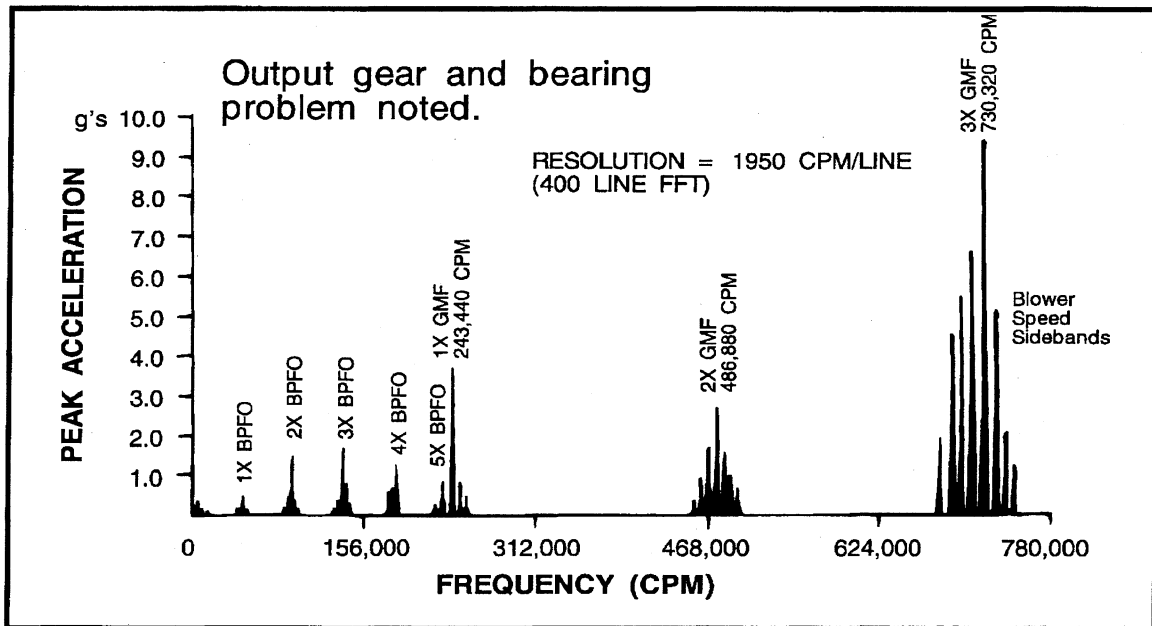
FIGURE 7



0-360,000 CPM SPECTRUM CAPTURES 1X GEAR MESH FREQUENCY IN ADDITION TO INFORMATION FROM FIGURE 6. RESOLUTION CONTINUES TO DECLINE.

FIGURE 8

Figure 9 shows data from the same measurement point as in Figures 5 thru 8, but F_{MAX} has again been increased up to 780,000 CPM. Also, note that the units have been changed to acceleration. This is because velocity response falls off in the higher frequency ranges where acceleration must be used. Note that we now see 2X GMF and 3X GMF. Also, note that 3X GMF is surrounded by a whole series of blower running speed sidebands which were not present in Figures 5 thru 8. In addition, note that 3X GMF is much higher than 1X GMF and dominates this acceleration spectrum. This machine must be attended to immediately with both a bearing and gear problem likely on the gearbox output shaft. However, it would not have been considered a serious problem machine if one had chosen an incorrect F_{MAX} such as that of Figures 5 and 6.



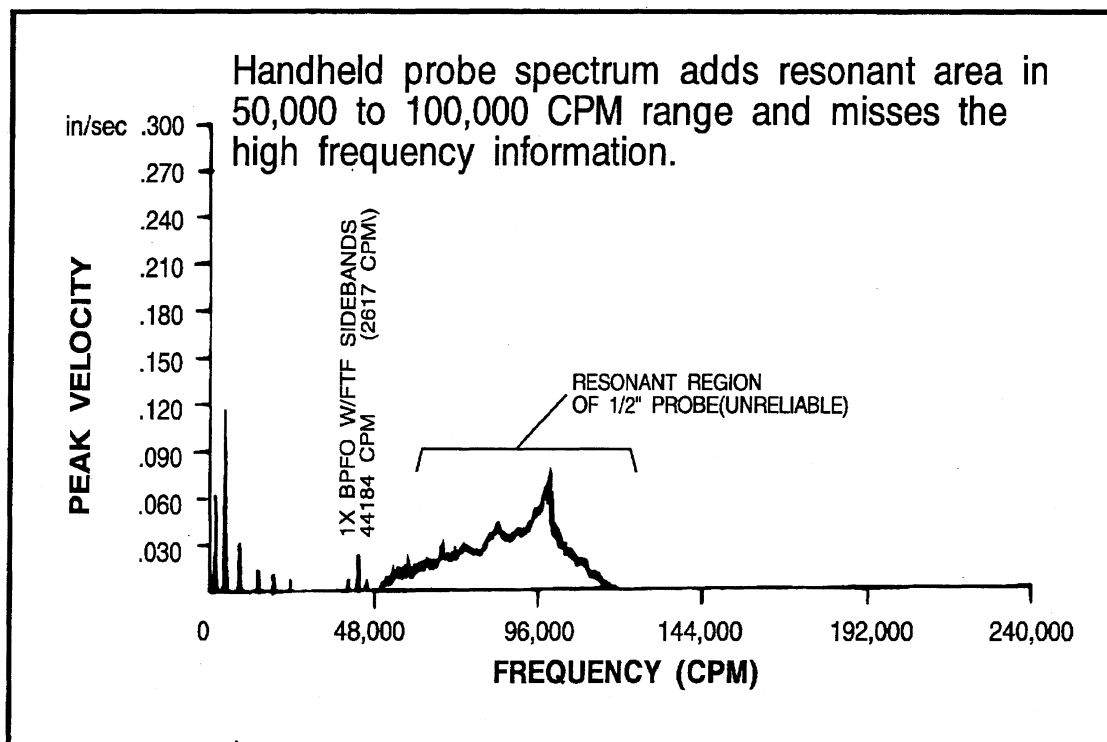
NOTE THE MEASUREMENT UNITS HAVE CHANGED FROM VELOCITY (in/sec) TO ACCELERATION (g's) DUE TO THE HIGH FREQUENCY RESOLUTION IS NOW SOMEWHAT POOR (1950 CPM), MEANING THAT 2 PEAKS MUST BE AT LEAST 2925 CPM APART FOR THEM TO EVEN BE DISPLAYED. A HIGH RESOLUTION SPECTRUM LIKELY USING 1600 LINES IS RECOMMENDED HERE. (Which would likely result in a resolution of 487.5 CPM - easily capable of displaying 1X RPM of motor and blower, along with clearly showing either bullgear (3580 RPM) or pinion speed (6086 RPM) sidebands if they are present).

0-780,000 CPM SPECTRUM CAPTURES HARMONICS OF GEAR MESH FREQUENCY IN ADDITION TO INFORMATION FROM FIGURE 8.

FIGURE 9

8.14 INSTRUMENT SELECTION, SETUP, AND CONDITION

1) Consider the allowable frequency range for both the data collector, transducer, and transducer mounting for each PMP machine. For example, if taking data on a low-speed agitator with a motor speed of 1780 RPM, but an agitator speed of only 60 RPM, it will be necessary to use both a low frequency data collector and a low frequency transducer on agitator measurement locations. Furthermore, transducer frequency range (governed on the high end by natural frequency) must be considered. Consider Figure 10 where the machine monitored in Figure 4 was again measured with an accelerometer outfitted with a 1/2 inch handheld probe having a natural frequency of approximately 96,000 CPM. Note that only the data up to approximately 48,000 is considered reliable in this case (with longer probes, this reliable frequency limit drops even lower). Above this frequency, however, the probe is becoming resonant and actually contributing "false" vibration data in the range of approximately 50,000 to 100,000 CPM. In other words, the plot of Figure 7 shows this raised noise floor is not even present. Furthermore, once well above the probe mount natural frequency (over approximately 120,000 CPM for this 1/2 inch probe) there is little or no response to vibration that in fact does exist. Therefore, machine problems that are present at these higher frequencies, as shown in Figures 7 thru 9, are no longer reported resulting in a false sense of security and can lead to unexpected catastrophic failure. Each probe and transducer has its own natural frequency and response curve which should be supplied when it is purchased. Be careful to fully understand exactly what that information is telling you.



HAND-HELD PROBE SPECTRUM AS COMPARED TO FIGURE 7 SPECTRUM.
FIGURE 10

- 2) Ensure the data collector batteries are in good condition and properly charged before beginning the measurement route.
- 3) Make sure cables are in good operating condition. Check them out before beginning the collection route.
- 4) Check for correct date and time in the data collector before taking data. This will affect both the overall measurement trends and spectral waterfalls since the date and time are "tagged" to this data in most PMP data collector/software systems.
- 5) Make sure the proper transducer sensitivity is entered into the data collector (as well as into the PMP software from each point if the transducer sensitivity is requested in the software point setup).
- 6) If additional equipment such as a temperature probe, current probe, phototach, proximity probe, or pressure transducers are used, ensure that their operation is clearly understood and the vibration instrument is properly configured to accept the information from them and connected to the proper input of the data collector. Ensure that the proper sensitivity is entered for the transducer (mV/psi for a pressure transducer; mV/amp for a current transformer, etc.)
- 7) Insure the transducer and/or other accessories are in good condition and are attached properly to the data collector. Check the connecting cable for proper operation, not only before, but periodically throughout the measurement route.
- 8) Be aware of instrument error stackup that can be as low as -14.5% and as high as 31.5%. The following excerpt from an article by John Catlin of IRD Mechanalysis (ref. 1) explains this:

The following table and text is reprinted from that article which indicates the amount of error that can be introduced by each particular piece of instrumentation or action:

	<u>Lowest Variation</u>	<u>Highest Variation</u>
Analyzer	<u>+5%</u>	<u>+10%</u>
Transducer	<u>+5%</u>	<u>+10%</u>
Magnetic Holder	<u>+0.5%</u>	<u>+0.5%</u>
Tape Recorder	<u>+1%</u>	<u>+2%</u>
Cable (Negligible if proper precautions are taken)		
Calibration	<u>+1.5%</u>	<u>+4%</u>
Attachment	<u>+1.5%</u>	<u>+5%</u>

"NOTE: The variations caused by factors of "transducer position", machine "environment", and "steady state" operation are not included in the above list because of the difficulty in establishing "lowest" and "highest" variation limits.

If the variations listed above are combined to give total maximum variations, the "Lowest Variation" range is -13% to +14%, while the "Highest Variation" range is -28% to +35%. The addition of "transducer position", "environment", and "steady state" variations can cause these percentages to increase significantly.

From the above, three conclusions can be drawn which should be helpful in carrying out a predictive maintenance program.

1) Do not attempt to make judgments on a machine's condition from small changes in amplitude which could readily be caused by variability of the instrumentation, transducer attachment, or normal variations associated with the environment and "steady state" operation of the machine.

2) Try to minimize the variability of the measurements by standardizing on one type measurement system and careful measurement (e.g., transducer attachment) procedures; but don't become so concerned with accuracy that the measurement procedure becomes excessively tedious and lengthy. Remember that small machinery defects generally continue to perform their required function even when balance levels increase by a factor of 2 to 10, or rolling-element bearing levels increase by a factor of 5 to 50. Thus, in most cases it is not necessary to make decisions on machine shutdown and repair based upon small changes in vibration amplitude.

3) Learn to recognize those machines whose vibration signatures have significant variations under "steady state" conditions because of the random nature of the inherent exciting vibrations. Pumps and fans which move liquids and gases are examples of machines which can fall in this category. In measuring such machines, avoid making decisions on machine condition based on changes which fall within the range of amplitude variations normally experienced during "steady state" operation. Alternately, make use of averaging to minimize the effects of the random variations on the measured amplitudes.

In summary, when carrying out a predictive maintenance program try to retain the same system components when making measurements. If it is necessary to change components (e.g. analyzer, transducer, etc.), run a check to determine the effect of such change on the total system response; and take any observed variations into account when interpreting the measurement results. Use careful measurement procedures to minimize variations caused by such sources; and be aware that many machines, even when running under "steady state" conditions, can have significant variations in their vibration signatures."

8.15 MEASUREMENT TECHNIQUES

1) Do not allow the transducer to rock back and forth, or to slip while taking measurements. If it should happen to do so, repeat the measurement (or the results can be very inaccurate and unreliable).

2) Apply firm continuous hand pressure to any hand-held measurement.

3) Be aware that magnetism and radio frequency (RF) waves can affect data collectors and/or transducers. If measuring in areas so affected, take special precautions such that both the analyzer and transducer are not effected.

4) Do not oversize a magnet and/or transducer with respect to the measurement location. This may affect or even change the vibration response if they are too massive, or if the magnetic field is strong enough to affect operation of the machine.

5) For low frequency measurements below 300 CPM, make sure the cable is not swinging back and forth. This motion can sometimes actually become part of the overall level or even part of the spectrum, and yet it has nothing at all to do with the machine itself.

6) Allow the vibration signal some time to settle before storing the measurement (normally 5 or 6 seconds unless taking data with a special low frequency transducer). Again, not doing so can cause very erroneous overall level and spectral results.

7) When at all possible, have vibration amplitude in the upper one-third of the measurement scale on those data collectors which allow the user to view the overall level before storing it and initiating the spectral measurement. In some data collectors, the full-scale amplitude of the spectrum will be set by the overall level. This can detrimentally affect the dynamic range if too high a full-scale amplitude is chosen (for example, if one chose a full-scale of 1.0 in/sec, and the amplitude of the highest peak was only .12 in/sec, the resulting spectrum may or may not pick up bearing defect frequencies with amplitudes of only about 0.010 to 0.015 in/sec).

8) Use the analysis and inspection features of data collectors if particular problems are noted and the additional information would be helpful. Examples of common inspection notes are: "leaking oil", "hot bearing", "loose belt", "machine down", "gravelly noise", etc.

8.16 TRANSDUCER MOUNTING AND PROBES

1) Use a magnet or quick-lock mount for taking data wherever possible. A "quick-lock" mount is a permanently mounted mechanism which allows certain transducers to be connected to it for the measurement by only a quarter turn of the transducer. It usually produces even better results than does the magnet mount.

2) Do not use a probe unless absolutely necessary. Even short 1/2 inch long probes can produce very unreliable results particularly if real frequencies exist above approximately 45,000 to 60,000 (750 to 1000 Hz) as was shown in Section 8.13. **Figure 11** shows 3 different spectra taken from the identical point on an induction motor driving a centrifugal air compressor. The only difference was the transducer mount itself. This was a 2-pole motor running at approximately 3563 RPM. This motor was outfitted with 45 rotor bars. Therefore, the Rotor Bar Passing Frequency (RBPF) was approximately 160,335 CPM (45 bars x 3563 RPM = 160,355 CPM). Here, it is important to point out that the same accelerometer and analyzer were used with all 3 measurements taken within a five minute interval. Looking at the spectra of Figure 11, note that the "magnet mount" spectrum at the top of the figure easily detected this Rotor Bar Passing Frequency (RPBF) at about 160,312 CPM (importantly, note that RBPF was by sideband frequencies spaced at exactly 7200 CPM which is 2X electrical line frequency). This is a common phenomenon in that rotor bar looseness is characterized by a high amplitude RBPF surrounded by 2X line frequency sidebands. On the other hand, the handheld measurement in the middle of the figure nearly missed RBPF altogether. And, even worse, when the 9" probe was attached to the accelerometer and the measurement repeated, not only did the probe mount totally miss the RBPF problem, but it also greatly amplified frequencies in the region of 40,000 to 60,000 CPM. Therefore, this figure is proof that a handheld probe should only be used when absolutely necessary since it will not only greatly amplify frequencies in one region, but will entirely miss important high frequency problems altogether.

Another example of the unreliability of probes or handheld mounts is shown in **Figure 12**. In this figure, an IRD 970 accelerometer was used with four different mounting techniques:

- A) Glue
- B) Magnet
- C) Handheld Only
- D) 9" Probe

Figure 12 - (A) is the most reliable data since it was collected with an adhesive mount. As can be seen, there is a bearing problem since the fundamental bearing defect and its harmonics are present.

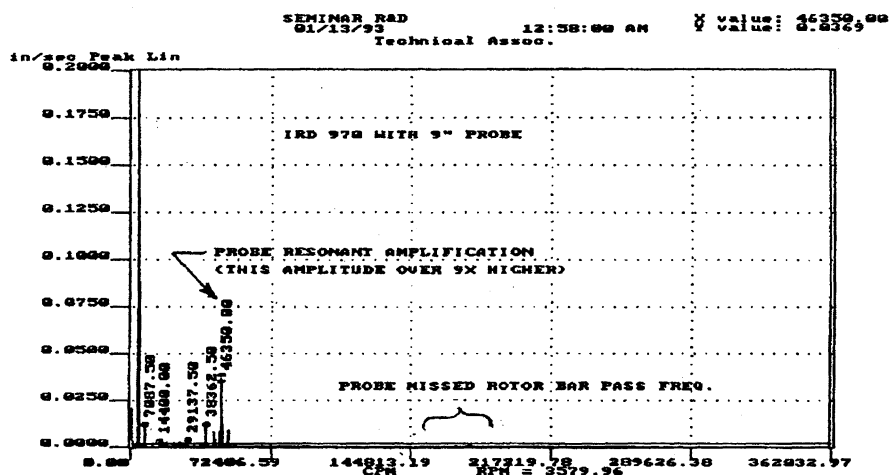
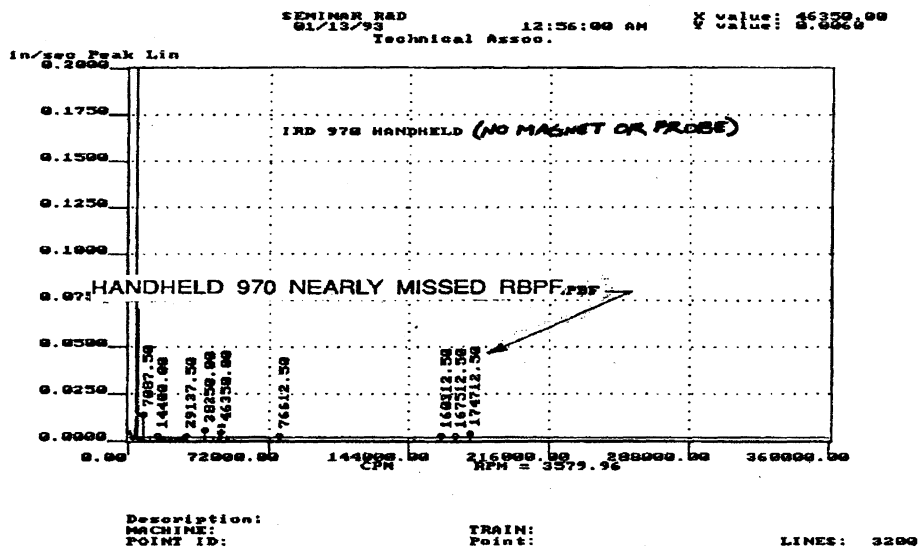
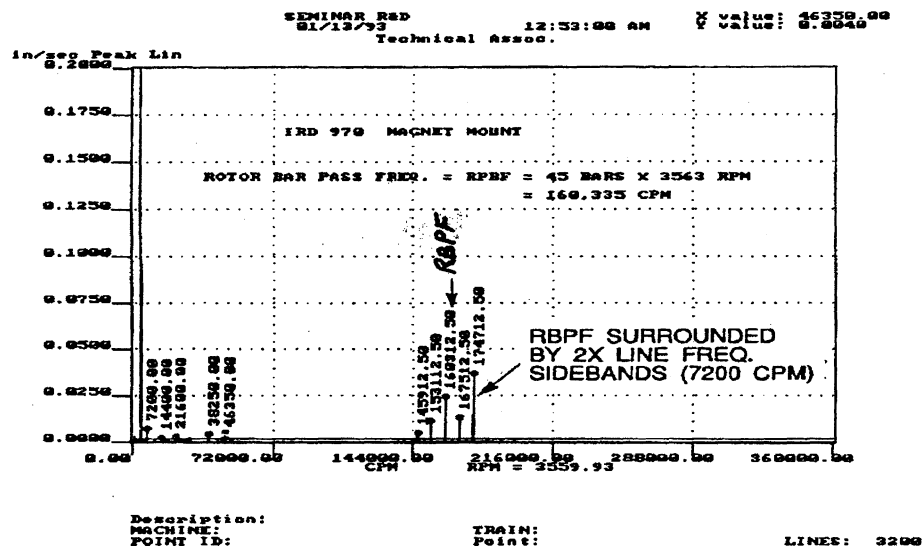


FIGURE 11
EFFECT OF TRANSDUCER MOUNT
ON RESULTING VIBRATION SPECTRUM

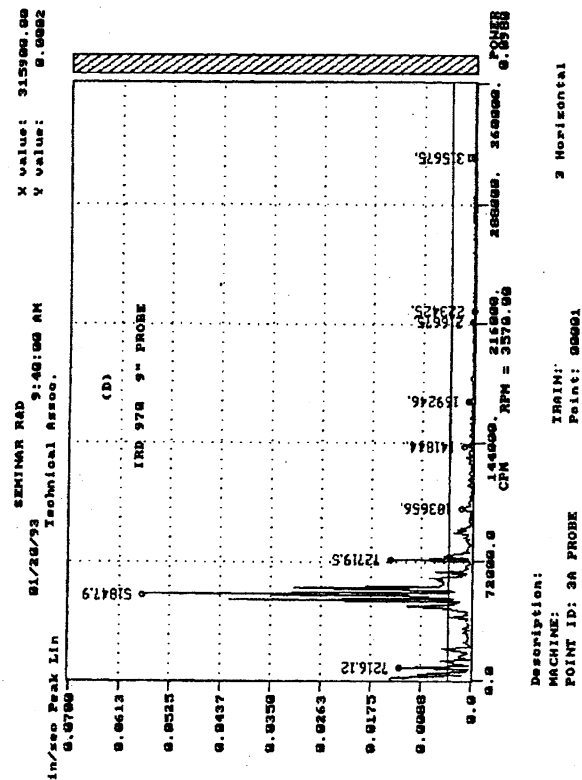
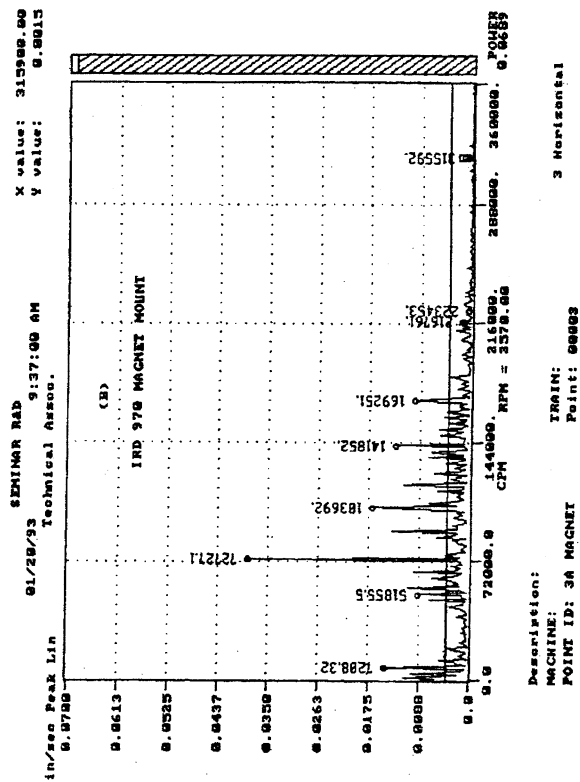
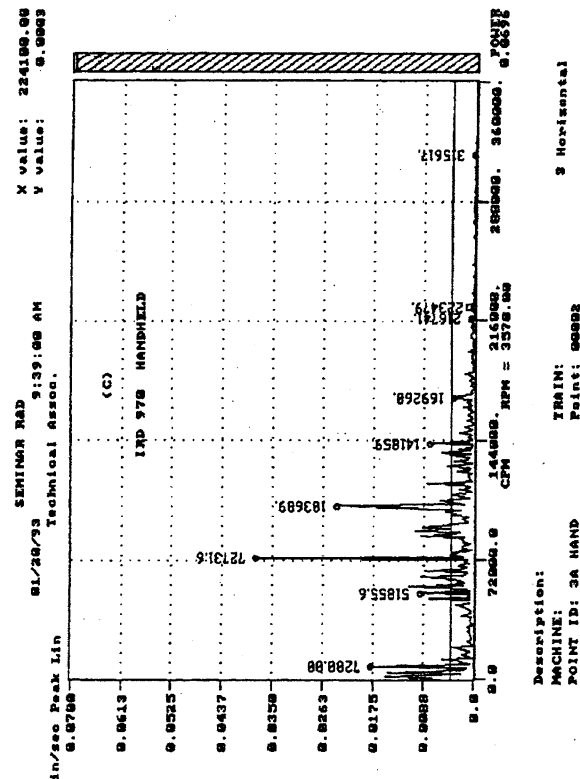
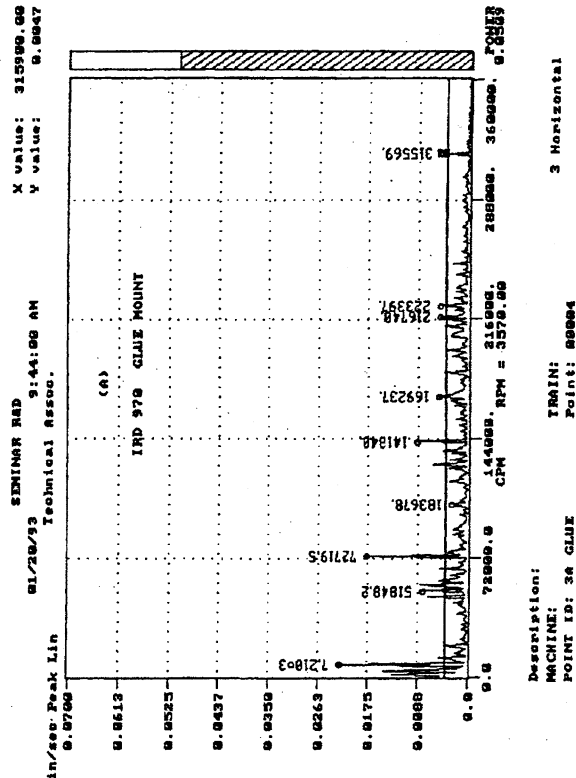


FIGURE 12

Figure 12 - (B) is the next best data since it was collected with a magnet mount. However, it is becoming more difficult to see the frequencies over 200,000 CPM since the magnet mount is not sufficient to "pick-up" those higher frequencies.

Figure 12 - (C) was collected with a handheld mount (no probe or magnet) and it is unacceptable. The bearing frequency harmonics no longer show up as peaks and this makes diagnosis on the condition of this bearing impossible. These peaks will never appear, even though they are present as can be seen in **Figure 12 - (A)**, since a handheld accelerometer is incapable of recording these high frequencies.

Figure 12 - (D) is the worst of all since it uses a 9" probe mount. This mounting procedure is unacceptable also. Not only are bearing frequencies missing but, the frequencies in the range of 40,000 to 60,000 CPM have been amplified greatly due to the resonance of the 9" probe. The 9" probe is only recommended in areas that are inaccessible or unsafe. These points should have permanent accelerometers mounted instead of using the probe.

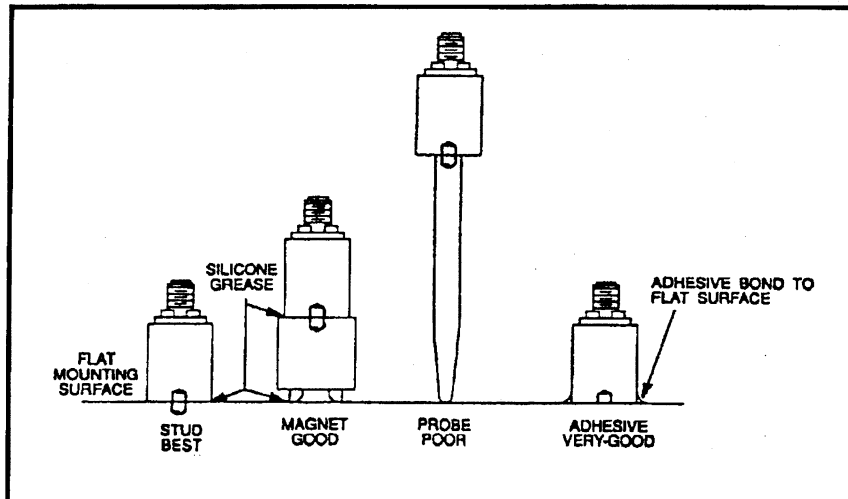
- 3) Do not take measurements on curved surfaces without using special precautions or a barred magnet. Doing so can greatly reduce the effective accurate frequency range of the transducer/magnet system.
- 4) Ensure contacting surfaces between the magnet and transducer are tight and free of dirt, grease, or other contamination. In addition, it likely will be necessary to scrape off the paint from the measurement location area if frequencies exceeding approximately 2000 Hz (120,000 CPM) are to be measured. Not doing so can significantly dampen or even eliminate the response from these high frequencies.
- 5) For the same reason as in Item 4 of this section, if mounting permanent washers or discs to a machine to provide a good measurement surface for the transducer, keep the thickness of the adhesive thin to allow good transmission of high frequencies through the adhesive itself. Also, ensure that the adhesive used has been tested and proven capable of freely passing these high frequencies (some adhesives themselves noticeably dampen such signals). Finally, if the permanent washer surface is uneven or rough, make sure that an adhesive is used which has the capability of filling these voids to provide a solid path for high frequency signal sources.
- 6) Ensure the transducer and/or magnet are held against a clean, dirt-free surface when taking measurements.

Pickup Mounting Methods

It is especially important that you meet certain requirements for ensuring the accuracy and repeatability of Spike Energy measurements. The technique you use for mounting the pickup is particularly important in this regard. Illustration A illustrates various mounting methods.

Stud mounting (or attachment with a high modulus adhesive bond) provides an effective method to attach an accelerometer to the machine surface. The primary advantage of this technique is that it provides the best path for the transmission of low level, high-frequency vibration from the machine to the pickup. It is, however, generally not suitable for periodic checks because attachment cannot be made quickly.

Another method uses a magnetic holder. Although a magnetic holder is simpler to use for periodic checks, there will be some loss of vibration energy between the machine and accelerometer. You can thus expect lower Spike Energy measurements than you would find with a stud-mounted accelerometer in the same position. For repeatability, care must be taken to locate the magnetic holder at precisely the same position each time measurements are made.



**MOUNTING METHODS FOR SPIKE ENERGY PICKUPS
ILLUSTRATION - A**

The third method uses a hand-held probe. The Spike Energy readings are normally lower than you would find for any of the other mounting techniques. This is mainly due to the attenuation of the probe. Again, repeatability of readings on a periodic-check program is dependent on the consistency with which you place the probe. The hand-held probe has the advantage that it can often be located closer to the bearing (tip contact) than when using the other mounting methods.

NOTE:

Do not compare Spike Energy measurements from the same point unless they are taken under the same conditions, including the mounting method.

It is important that you attach the accelerometer in exactly the same place each time a measurement is made. A circle painted around a mounted accelerometer makes it easier to reattach it in the proper position for future measurements.

The surface where the measurement is to be taken must be kept clean and bare. Surface materials such as paint or grease can act as a spring under the accelerometer, thus lowering its resonant frequency and reducing high-frequency Spike Energy amplitudes. Also, the surface should be flat so that the accelerometer does not wobble when attached. When using a stud mounting, its axis must be perpendicular to the surface so that the accelerometer sets squarely on top of it.

Use a light coating of silicon grease or lubricating oil between the accelerometer and mounting surface. This forms a stiff, incompressible layer that transmits high-frequency vibration. When using a magnetic holder, applying a light coating of oil to the mounting surface improves high-frequency transmission.

8.2 EFFECT OF TRANSDUCER MOUNTING ON VIBRATION MEASUREMENTS

The following is an article by CSI (ref. 2) which outlines the effects of transducer mounting on the usable frequency range. The transducer mountings studied are stud, adhesive, quick-lock and magnetic. It is considered one of the better studies on this important topic published to date. Note particularly what is said about each of the mount configurations (esp. the information on hand-held probes versus magnets or quick-lock mounts). The article also provides good information on what adhesives should be used when mounting disks or washers permanently.

Real-World Mounting of Accelerometers for Machinery Monitoring

S. V. Bowers, K. R. Piety and R. W. Piety, Computational Systems Incorporated, Knoxville, Tennessee

Typical accelerometer mounting configurations were evaluated to determine acceptable frequency response and resonant frequencies for each mounting. Basic configurations studied included stud, adhesive, quick lock and magnetic. A hand held probe with varying stinger lengths was also evaluated. In addition, the effects of bad mounting techniques were considered. Frequency ranges and resonances were determined for sensor mounts exposed to rocking, looseness and varying hand pressure.

Predictive maintenance programs that operate in utilities and major industries rely heavily on manual data collection. The temporary sensor mounting techniques utilized in manual data acquisition significantly impact data repeatability and reliability and thus the decisions based on these data. The benefits of these monitoring programs are dependent upon accurate data collection techniques which affect both detection and diagnosis. Provided vibration instrumentation is calibrated, accurate vibration measurements are ensured when: (1) sensor mountings do not limit frequency and dynamic ranges; (2) sensor mass does not change the vibration characteristics of the test object; and (3) measurement locations are exactly repeated.¹

The objective of this project was to evaluate the limitations of some common temporary sensor mounting techniques as they apply to predictive maintenance programs. This was achieved in three major sets of experiments. For the first set, maximum frequency responses were determined for common sensor mount configurations. In the second set of tests, misapplication of sensor mounts was investigated. The third set of experiments explored the effects of several sources of variability on vibration measurements.

Frequency response limits for a given sensor are defined as the linear range of the frequency response curve for that sensor. It is desirable for the output from the sensor to be linearly related to the input over as wide a frequency range and dynamic range as possible.¹ A main focus of this study was to determine sensor frequency response limits resulting from various mount configurations. In practical applications, the actual frequency response of a sensor depends upon the dynamic response of the sensor/mount configuration. Vibrations at frequencies near the sensor/mount configuration resonant frequency will be amplified, whereas at other frequencies the measured vibration is greatly attenuated. The frequency ranges and resonant frequencies of various mount fixtures are reported in this article.

The usable frequency range of a mounted sensor can be reduced by rocking and loose connections. In addition to these factors, the frequency range for hand held probes can be reduced due to variations in hand pressure. Limitations imposed by these misapplications were evaluated. Because of improper mounting and irregular or dirty surfaces, rocking occurs, which tends to decrease the usable frequency range of the transducer.² Looseness occurs when mount configurations and sensor are not tightly connected. This problem results in large errors at high frequencies,³ as well as chattering. Hand held probes can also experience reductions in frequency response due to chattering because of improper hand pressure.³ Any action that allows the pickup axis to change direc-

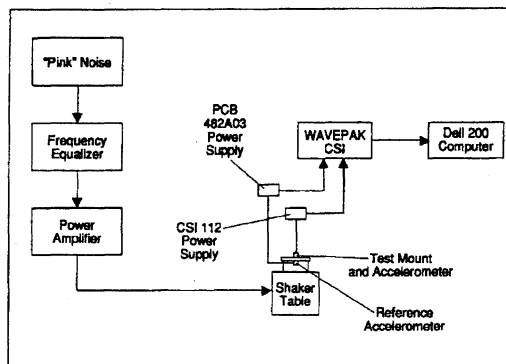


Figure 1. Schematic of instrumentation to evaluate sensor mount configurations.

tion also produces incorrect readings.

Other sources contributing to variability in measured results are the mount positions on vibrating surfaces, varying sensor placement techniques by different people, and individual consistency. This article presents results relating to these factors and attempts to quantify their effects on measured values in typical or extreme cases. Tests were evaluated on industrial machines from data obtained with an accelerometer and a hand held probe using common mount configurations currently employed in industrial predictive maintenance programs.

Sensor Mount Configurations

During the sensor response tests, accelerometer outputs were measured using a CSI Wavepak which is a two-channel FFT spectrum analyzer integrated with a personal computer. Each measurement, unless otherwise noted, was taken with 800 lines of resolution and 200 averages. The instrumentation employed in this study is indicated in the block diagram of Figure 1. "Pink" noise was generated and passed through a frequency equalizer to provide a relatively constant output per unit frequency between 0 and 20,000 Hz. Output of the power amplifier was adjusted to insure an overall RMS value of 1 g in the measured spectra.

Test data were comprised of transfer function, phase and coherence curves determined from test and reference accelerometer measurements of each mount configuration. The reference accelerometer (Wilcoxon 726) was always stud mounted on the bottom of the plate and torqued to 20 in-lbs. All test accelerometer measurements were taken from the top of the plate. With the exception of the hand held accelerometer tests, experiments were conducted using a Wilcoxon 726T accelerometer as the test accelerometer. The manufacturer rated the frequency response of the 726 accelerometer at 3 to 10,000 Hz ($\pm 5\%$). When stud mounted, the resonant frequency for this accelerometer is 35,000 Hz. The rated frequency response of the CSI 310 hand held probe stud mounted and tightened to a vibrating surface by hand is 2 to 2000 Hz.

Test configuration data were normalized by base data in order to reduce or remove measurement bias resulting from the test procedures employed. Normalization requires complex division of the test data by the base data. The normalizing process was considered valid provided both the test and base data had a coherence of 0.95 or above. Base data were taken

Based on a paper presented at the 4th Incipient Failure Detection Conference, Electric Power Research Institute, Philadelphia, PA, October, 1990.

periodically throughout the study to verify the consistency of test conditions and results. The base data were acquired with the test accelerometer (Wilcoxon 726T) stud mounted to the top of the steel test fixture plate to a torque of 15 in-lbs and the reference accelerometer stud mounted to the bottom of this test fixture plate to a torque of 20 in-lbs.

By normalizing the test data, a transfer function ratio, minimum coherence curve and phase difference were determined. Any transfer function ratio deviation exceeding 0.90 and 1.10 (greater than $\pm 10\%$) was considered significant as was a phase difference greater than $\pm 10^\circ$. These limits were selected for the purpose of establishing a recommended linear range of application for each specific mount configuration. Resonant frequencies of each mount configuration were also determined when possible. The resonant frequency of a configuration was noted when a transfer function ratio peak occurred at a phase difference of approximately 90° .

Results, Discussion and Conclusions. Basic mount configurations considered were: 1. stud mount, 2. adhesive mount, 3. quick-lock mount, 4. magnet mount, and 5. hand held probe. For the most consistent and reliable data, it is best that readings from any given measurement point be taken from smooth surfaces where the path from the sensor to the monitored point is direct. This is best accomplished by stud mounting a steel disk at desired measurement points. The only requirement for stud mounting a disk to a machine is that the machine surface be smooth and flat. A 1 in. diameter, 3/8 in. thick steel disk was stud mounted to the vibrating surface in this study. The frequency range of the accelerometer stud mounted to this fixture was 3 to 16,125 Hz as dictated by the phase difference boundary. This range is easily large enough for practical applications. For cases where the mounting surface is round, the disk may be applied using certain cement adhesives. Evaluation of a good adhesive for such a use will be discussed shortly.

When mounting disks, or any sensor mount, it is important that the machine surface be smooth especially when higher frequency measurements are desired. Voids between the measured surface and the sensor will reduce the accelerometer's frequency range. This is best demonstrated by comparing experiments conducted with and without wax. After applying wax to a fixture, frequency ranges were enhanced by as little as 925 Hz to as much as 4,750 Hz. The frequency ranges were improved because voids between the mount fixture and vibrating surfaces were filled allowing better transmission between contacting surfaces. In practical measurement situations, applying wax is not feasible due to surrounding debris and excessive temperatures. However in situations where higher frequencies must be analyzed, wax can be applied provided the environment is relatively clean and the temperature is below 200°F (wax melts above 200°F).

As alluded to previously, steel disks can also be bonded to machines via adhesives. The only requirements are that the surface be clean and smooth (no drills or taps needed). Mounting procedures for the tests in this study were identical to those required in industrial applications. Each adhesive was applied using the procedures described by the manufacturer of the particular bonding agent.

The accelerometer's frequency range resulting from the configuration using the Hottinger Baldwin Messtechnik X60 adhesive is 3 to 8,775 Hz. When the setup used the Loctite Black Max cyanoacrylate ester adhesive, the accelerometer's frequency range was 3 to 10,975 Hz. No resonance was observed for either of the adhesive mount configurations. Of the two bonding agents, Loctite Black Max was easier to apply. However, this adhesive will only bond surfaces that are flat and smooth. In situations where one or both of the mating surfaces is rounded (not flat), Hottinger Baldwin is ideal.

Sometimes, sensors must be isolated from mount configurations. In such situations, electrical insulators are placed between the sensor and mounting interface. The use of insulators tends to lower the maximum acceptable frequency. Three common insulators were evaluated in this study. The maxi-

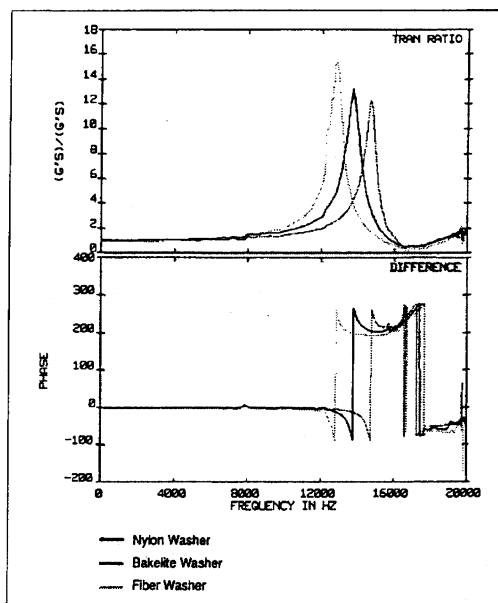


Figure 2. Transfer function ratios (top graph) and phase relationships (bottom graph) resulting from stud mounting of a Wilcoxon 726T accelerometer with three different insulated washers.

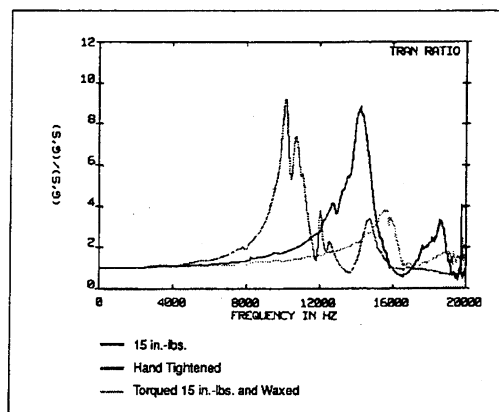


Figure 3. Transfer function ratios for three different quick lock configurations.

mum acceptable frequency of the stud mounted accelerometer with insulators ranged from 4200 to 5400 Hz. The transfer function ratios and phase difference curves for these three tests are plotted in Figure 2.

One mounting system used in lieu of the steel disk is the quick lock. This system consists of a base permanently attached to a measurement point and a cap connected to the sensor. The cap is easily coupled to the base with 3/4 of a turn. This connection is usually tightened by hand. Results of a test where the base was stud mounted at 15 in-lbs to the vibrating surface and the remainder of the quick lock system and sensor were hand tightened to the base show an acceptable frequency response ranging from 3 to about 4,000 Hz (see Figure 3). While a test was not conducted with the quick lock base cemented to a vibrating surface, the expected frequency response of such a configuration would be the same as the stud mounted quick lock system, i.e. 3 to 4,200 Hz. The frequency range of the quick lock configuration is limited because of the quick lock device, not the method of permanent connection to the vibration surface.

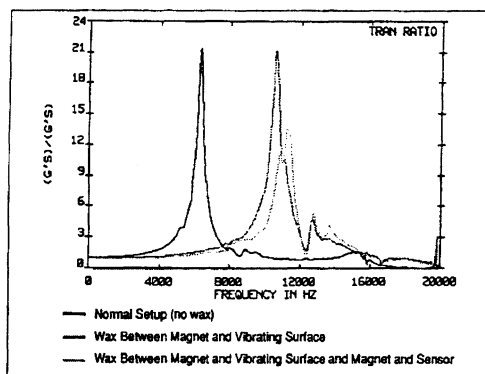


Figure 4. Transfer function ratios for three configurations of a Wilcoxon 726T accelerometer mounted on a Balmac magnet.

In many situations, permanent vibration mounting disks are not feasible because of accessibility and location of measurement points. The best solution in this case is to mount sensors on magnets. Three different magnets were evaluated in the experiments. For each mount, the accelerometer was stud mounted at 15 in-lbs to the magnet being tested. The frequency range of the accelerometer mounted on a rare earth magnet was the widest and ranged from 3 to 3,350 Hz. The frequency response of the Balmac magnet and CSI 906 super magnet were similar with ranges from 3 to 1,775 and 3 to 1,475 Hz, respectively. An example of transfer function ratios from the Balmac magnet configurations with and without wax are plotted in Figure 4. Even though the rare earth magnet had a better response, this magnet can only be used on flat surfaces because the face of the magnet is flat. The Balmac magnet is capable of holding onto flat and slightly rounded surfaces, while the design of the CSI 906 super magnet allows it to be used on practically any rounded or flat surface.

The final mount technique studied utilized a CSI 310 hand held probe and stinger configuration. Hand held probes should be used when measurement points are inaccessible by any other method. Typical stinger lengths are 2 and 8.5 inches. The maximum acceptable frequency for these configurations was much less than any of the other tests. For the hand held probe with a 2 in. stinger attachment, the maximum acceptable frequency was 525 Hz and for the 8.5 in. stinger attachment it was 350 Hz. It is not recommended that hand held probes be mounted on magnets because cantilever action can occur which will produce erroneous data.

The most common mount configurations of those evaluated are ranked in Table 1 with respect to the valid frequency response in decreasing order. The suggested frequency ranges and resonant frequencies are liberal estimates and will differ slightly under various conditions and for different sensors. The value observed for the recommended frequency range was influenced by the shape of the transfer function ratio curve for each mount. If the curve was smooth and sloping gently upward, the recommended value was chosen objectively at the base of the resonant curve where the graph was generally flat (linear relationship). Therefore, it is possible for the recommended maximum value to be larger than that observed.

Table 1 also shows that different mounting configurations tend to amplify measurements about their resonant frequencies to varying degrees. In general, hand held probe measurements amplify signals about the resonant frequency less (6x) than do the other configurations. One can expect a mounting system employing insulators to increase readings about the resonant frequency by about 12 to 16x the actual vibration. Magnetically mounted sensors will amplify vibration around the resonant frequency by 13x when using a rare earth magnet to over 20x when using a CSI 906 or Balmac magnet. Finally, when using a quick lock configuration, vibration can be ampli-

fied about the resonant frequency by 10x. These data suggest that measurements should not be taken at or around resonance when actual amplitudes are desired because the true vibration magnitudes will be amplified.

Poor Sensor Mounting Practices

Various misapplications associated with sensor mounting were investigated. The equipment and procedures used for tests in the previous section were applied to the experiments in this section. A description of the mount configurations tested is outlined in Tables 2 through 4. The maximum acceptable frequency with respect to the transfer function ratio and phase difference along with the resonant frequency for each mount are also recorded in Tables 2 through 4.

Results, Discussion and Conclusions. In most "real world" applications, the mounting configuration can be far from ideal. For example, rocking occurs when the vibrating surface and the mount or sensor do not fully and completely mate. Undesirable debris accumulation between the two surfaces tends to rock the sensor. To evaluate the severity of this problem, ideal and rocking modes were measured for stud mount and magnet mount configurations. Rocking was induced by randomly placing drill press residue between the test fixture plate and the contacting surface of several mounting configurations. Extra rocking was introduced into the magnetic mount by piling metal shavings in a row which created a "seesaw" effect.

The effect rocking has on a sensor's frequency range was apparent. Without the presence of rocking, earlier results revealed the frequency range for a basic stud mounted accelerometer to be 3 to 17,450 Hz. With rocking, the frequency

Table 1. Maximum acceptable measurement frequency limits and mounting resonances.

Sensor Mount Configuration*	Maximum Acceptable Frequency (Hz)**	Mounting Resonance Freq(Hz)	Ampl(g)
Sensor stud mounted to 3/8" steel plate stud mounted to vibrating surface	16,200	>20,000	N.O.
Sensor stud mounted to steel disk bonded to vibrating surface with Loctite Black Max cyanoacrylate ester	10,975	N.O.	N.O.
Sensor stud mounted to steel disk bonded to vibrating surface with Hottinger Baldwin Messtechnik X60	9,000	N.O.	N.O.
Sensor stud mounted with a:			
bakelite insulator	10,000	14,650	12.0
nylon insulator	9,000	13,750	13.5
fibre insulator	9,000	12,075	15.5
Sensor stud mounted to rare earth magnet	7,500	12,075	13.0
Sensor mounted to quick connect stud mounted to vibrating surface (hand tightened)	6,000	10,150	9.0
Sensor stud mounted to Balmac magnet	3,000	6,325	21.0
Sensor stud mounted to CSI 906 super magnet	2,000	5,250	22.0
CSI 310 probe using a pointed 2 in. steel stinger†	800	1,475	6.5
CSI 310 probe using a blunt 8.5 in. aluminum stinger†	500	1,075	6.0

* All stud mounts torqued at 15 in-lbs unless otherwise noted.

** Maximum acceptable frequency chosen objectively at the base of the transfer function curve where the slope of the line begins to increase towards the resonant peak but is still generally flat (linear relationship).

Freq Mounted resonance frequency.

Ampl Normalized amplitude at the mounted resonance frequency.

N.O. None Observed

† Stingers tightened to accelerometer at 15 in-lbs. Medium pressure applied to hand held probe.

ranges observed were 3 to 6000 Hz for the sensor torqued to 5 in.-lbs, 3 to 2150 Hz for the sensor finger tightened and only 3 to 775 Hz when the sensor was loose (Table 2). These results showed reductions in frequency ranges of 11,450, 15,300 and 16,675 Hz, respectively. Degradation in frequency response can be partially attributed to loose sensor mounting.

Tests conducted with magnet mounts revealed reductions in

Table 2. Evaluation of rocking sensor mounts.

Sensor Mount Configuration	Maximum Acceptable Frequency (Hz)	Mounting Resonance Frequency (Hz)
Metal shavings between contacting surfaces of the sensor, stud mounted to the vibration pad for:		
normal mount*	17,450	>20,000
5 in.-lbs torque	6,000	14,350
finger tightness	2,150	7,350
loose connection	775	2,725
Sensor stud mounted to rare earth magnet:		
normal mount*	2,975	8,875
metal shavings between magnet and vibration pad	1,800	6,125
metal shavings piled to one side	1,200	5,075
Sensor stud mounted to Balmac magnet:		
normal mount*	2,400	8,525
metal shavings between magnet and vibration pad	1,200	4,000
Sensor stud mounted to CSI 906 super magnet:		
normal mount*	1,250	3,550
metal shavings between magnet and vibration pad	750	2,650

*Torqued at 15 in.-lbs.

Table 3. Evaluation of loose sensor mounts.

Sensor Mount Configuration	Maximum Acceptable Frequency (Hz)	Mounting Resonance Frequency (Hz)
Sensor stud mounted to the vibration pad for:		
normal mount*	17,450	>20,000
loose mount	1,050	2,450
Sensor stud mounted to rare earth magnet:		
normal mount*	2,975	8,875
finger tight	2,200	6,725
Sensor stud mounted to Balmac magnet:		
normal mount*	2,400	8,525
loose mount	500	3,375
CSI 310 handheld probe using a pointed 2 in. steel stinger:		
medium pressure	425	1,210
loose stinger	385	1,045

*Torqued at 15 in.-lbs.

Table 4. Results of hand pressure applied to a hand held probe.

Sensor Mount Configuration	Maximum Acceptable Frequency (Hz)	Mounting Resonance Frequency (Hz)
CSI 310 hand held probe using a 0.25 in. stud as a stinger at pressures:		
light	395	1,050
medium	550	1,420
heavy	605	1,630
CSI 310 handheld probe using an 8.5 in. aluminum stinger at pressures:		
light	280	750
medium	320	875
heavy	415	1,070

*Torqued at 15 in.-lbs.

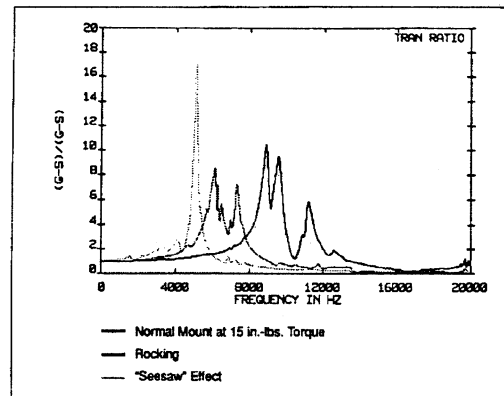


Figure 5. Comparisons with respect to the transfer function ratios of a Wilcoxon 726T accelerometer mounted on a rare earth magnet in the normal, rocking and seesaw modes.

frequency responses due to rocking as well. The effects of rocking reduced the maximum acceptable frequencies from as little as 500 Hz to as much as 1,775 Hz. An example showing the effects of rocking on a magnet mount is seen in the transfer function ratios of Figure 5.

Results suggest that measurement point surfaces and sensor interfaces be thoroughly cleaned and connections tightened in order to minimize the influence interface dynamics can have on data, particularly when data are being taken above 500 Hz. Furthermore, in a situation where a magnet is walking around on a surface or intermittently losing contact, low frequency motion and impacts (which are not generated inside the machine but due to a poor interface) will raise the noise floor of measured vibration spectra.

Mounting looseness occurs when any connection between the sensor and a vibrating machine is not securely tightened. Looseness often appears after moving from measurement point to measurement point without periodically checking the sensor and mount connection. Hand contact, vibrations from the equipment being monitored, and temperature fluctuations can eventually loosen sensors from their mounts. Looseness allows the sensor to vibrate randomly. This reduces the usable frequency range of the sensor and increases the probability of acquiring spurious data.

Two degrees of looseness were evaluated: "finger tight" and "loose mount." The finger tight mount was achieved by twisting the accelerometer to the mount with the fingers only. Thus, the sensor was snugly mounted, but not to the degree of hand tightness. The degree of looseness was defined as "loose mount" when movement between the sensor and the mount could be described as a random motion.

The usable frequency range measured for the "loosely" stud mounted accelerometer is 3 to 1050 Hz (Table 3), a reduction of 16,400 Hz from the basic stud mount setup. In the experiment where the sensor on a rare earth magnet was tightened with only the fingers, the frequency range was reduced by 775 Hz so that the adjusted frequency range was 3 to 2200 Hz. The frequency range with the accelerometer "loosely mounted" on a Balmac magnet was 4 to 500 Hz, a reduction of 1900 Hz to the regular Balmac magnet mount frequency range. The effect of this loose mount setup was further observed by evaluating the transfer function ratio and phase difference curve of Figure 6. The small peaks appearing at various frequencies indicate that the sensor was chattering. Because of chattering, high erroneous readings can be obtained. Such readings, especially at low frequencies, could mislead the analyst to conclude that mechanical problems are present.

With regard to the hand held probe, a loose connection produced the sensor frequency range of 3 to 385 Hz, a reduction in frequency response of 40 Hz relative to the preferred hand held

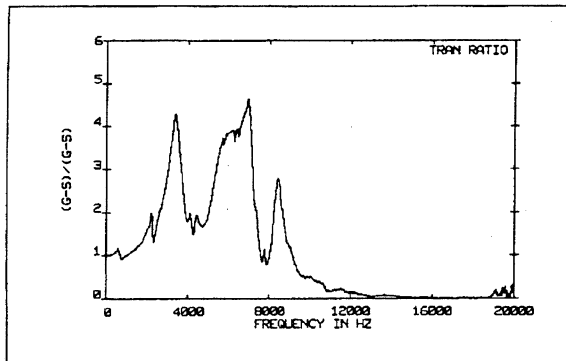


Figure 6. Transfer function ratio showing chattering produced by a loose connection between a Wilcoxon 726T accelerometer and a Balmac magnet.

probe configuration. While reductions due to mounting looseness were not as severe as found in other mount configurations, the fact that tests were performed in a clean environment had some effect. Even though the stinger attachment was loose, the pressure applied to the probe eliminated some of the problems normally caused by slack joints and chatter. In field conditions, the technician experiences less than ideal conditions and vibrations occur in the vertical, horizontal and axial directions (not just vertically). A loose stinger connection in a field environment is likely to deteriorate the output signal more than it did in our laboratory tests because of the additional vibrations present in the horizontal and axial directions.

The hand pressure applied to hand held probes will vary between different people and among individuals. Therefore, tests were conducted to study the effect force variations have on hand held probes. For this experiment, two stingers, each connected to a CSI 310 hand held probe, were evaluated at three different degrees of force: light, medium and heavy (see Table 4 for results). Of the two stingers, the first stinger considered was a 10-32 UNF stud threaded tightly into the sensor until it extended 0.25 in. beyond the probe's tip. The frequency ranges for the light, medium and heavy pressure measurements were 3 to 395 Hz, 3 to 550 Hz and 3 to 605 Hz, respectively. For each decrement of hand force with respect to the heavily applied pressure measurement, the frequency ranges decreased 55 and 210 Hz. This changing value will vary with increases and decreases in hand pressure. Figure 7 compares the transfer function ratios for each of the three hand pressures.

When using the 8.5 in. aluminum stinger attachment, the acceptable frequency response ranged from 3 to 280, 3 to 320 and 3 to 415 Hz for the data collected using the hand held probe with light, medium and heavy hand forces, respectively. The frequency range decreased 95 and 135 Hz with decrease in vertical force.

The results in Table 4 show that the resonant frequency and therefore frequency range of the hand held probe varies proportionally with hand pressure. For measuring vibrations in the upper portion of the sensor's frequency range, consistent hand pressure is essential. Otherwise, erroneous readings may be obtained.

Data Collection Inconsistencies

Variability can occur due to measurement location, various people collecting data and individual inconsistency. The potential for introducing variability in the measured data was evaluated on bearing housings of rotating machinery utilizing typical data collection methods. A sketch showing the locations of measurement points for the two bearing housings studied appears in Figure 8. By collecting cross spectral data, all test data from each bearing housing could be compared and normalized with respect to its base data. The reference accelerometer used on the first bearing housing, which had a rounded surface, was a Wilcoxon 726T stud mounted (15 in-

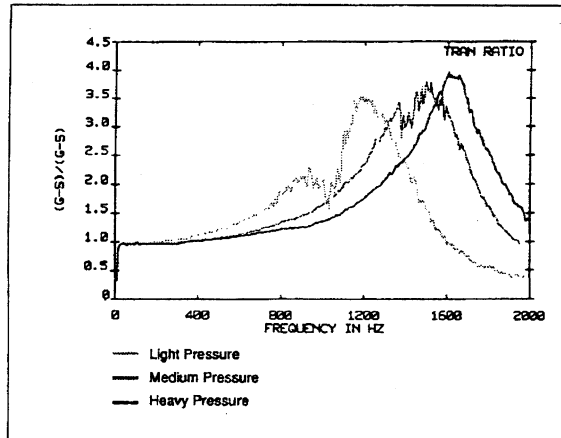


Figure 7. Comparisons, with respect to transfer function ratios, of the effects of hand pressure applied to a CSI 310 hand held probe with a 0.25 in. steel stinger attachment.

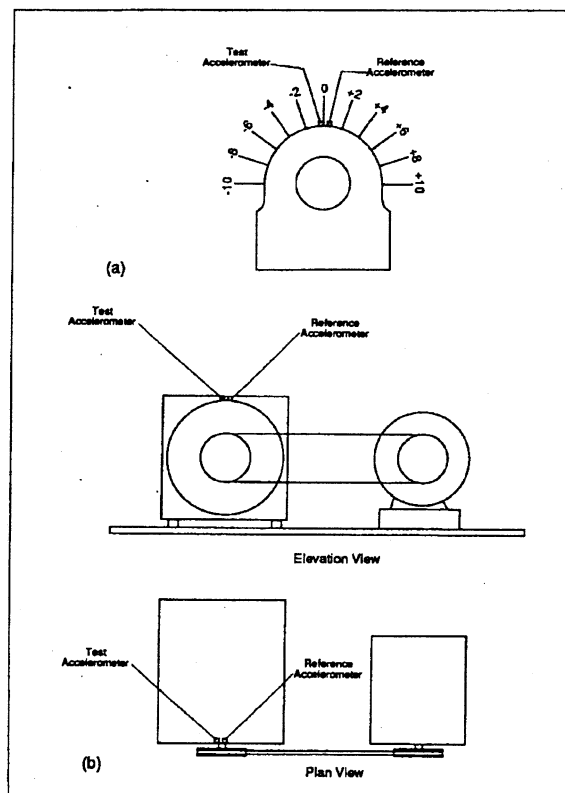


Figure 8. Schematic of: (a) bearing housing number 1 (numbers correspond to in. from the reference point 0); and (b) industrial machine driven by a DC motor (bearing housing number 2). Measurement points are indicated in each drawing.

lbs) to a CSI 906 super magnet and placed at position 0 as seen in Figure 8a. For measurements conducted on bearing housing number 2, which had a flat surface for mounting, the reference accelerometer (Wilcoxon 726T) was mounted with a rare earth magnet at the position indicated in Figure 8b.

Variability Due to Measurement Location. Data variability due to measurement point location was evaluated on bearing housing number 1 which contained a shaft turning at a speed of approximately 4 Hz. A Wilcoxon 726T accelerometer, also mounted on a CSI 906 super magnet, was used to collect the test measurements. A single spectrum was obtained at each

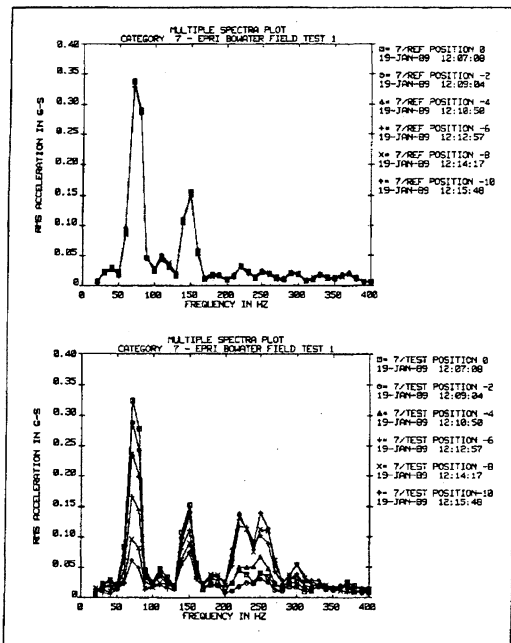


Figure 9. Spectra ranging from 0 to 400 Hz for reference (top graph) and test measurements (bottom graph) corresponding to data taken from the negative side of the bearing housing shown in Figure 8a.

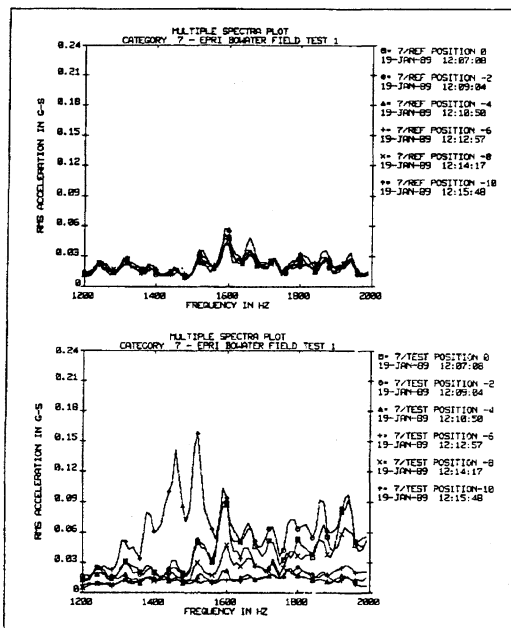


Figure 10. Spectra ranging from 1200 to 2000 Hz for reference (top graph) and test measurements (bottom graph) corresponding to data taken from the negative side of the bearing housing shown in Figure 8a.

measurement point over a frequency range of 0 to 2000 Hz by collecting twenty-five averages with 200 lines of resolution.

Comparisons of spectra computed from reference and test accelerometer measurements, taken from the negative side of bearing housing number 1, are displayed over the range of 0 to 400 Hz in Figure 9 and 1200 to 2000 Hz in Figure 10. Spectra (not shown) produced from the reference and test accelerometer readings were acquired from the positive side of the bearing

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housing yielding similar results. The frequency range between 400 and 1200 Hz is not displayed because data in this range were small in amplitude relative to the other frequencies.

Results showed that the magnitude for any given frequency was different for each change in measurement location. The largest magnitude at any given frequency occurred in the vertical and/or horizontal positions; i.e. some vibrations proved to be prominent vertically, while others were predominant horizontally. It was noted that a mere 2 in. displacement of the sensor on a housing approximately 40 in. in circumference would result in changed magnitudes ranging from 0.001 to 0.3 g depending on the frequency range evaluated. On other machines, the output change depends on the size of the vibrating surface. Larger surfaces may allow for more liberal movement of sensors while small machines may require a more specific measurement location in order to avoid collecting invalid data.

Measurement points should be chosen and marked in the vertical, horizontal and/or axial directions. The best marking method would be to mount (stud or adhesive) a washer or mounting disk onto the machine surface. Not only will these mounts be permanent, but the measuring surface will be flat. Another method that will improve the reliability of data is to mark the machine housing for each measurement location with a spot of paint or indentation. In general, the more time and money put into marking measurement points, the more reliable the resulting data. One must be careful when choosing the position for measurement on a housing surface. Measurement points should be at positions having solid metal paths from the sensor to the source of vibration. Interfaces and internal voids can attenuate vibration and provide unreliable data.

Variability Resulting in Measurements by Different People. The variability in vibration readings taken by different people and the variability in repeated measurements by the same individual were evaluated on bearing housing number 2 which was running at a speed of 1440 rpm (24 Hz). All measurements were taken from the same marked point on the bearing housing in order to eliminate changes in measurement location as a factor contributing to the variability of the data. Base data were obtained using Wilcoxon 726T accelerometers. Both the reference and test accelerometer were stud mounted (torque of 15 in-lbs) on rare earth magnets.

All measurements taken from the bearing housing in Figure 8b were evaluated by calculating differences in magnitude of the reference accelerometer minus magnitude of the test accelerometer for prominent frequency ranges of concern. By comparing differences, the variance due to motor speed and reference sensor location was eliminated.

Variability introduced when different people collect data was assessed on the basis of the two tests described below:

1. Five individuals acquired acceleration measurements using a hand held probe (CSI 310) with a pointed 2 in. steel stinger attachment. Spectra with a frequency range of 0 to 2000 Hz were obtained with six averages and 400 lines of resolution.
2. Five individuals took acceleration measurements with the same Wilcoxon 726T accelerometer mounted on a rare earth magnet. Spectra with a frequency range of 0 to 2000 Hz were obtained with six averages and 400 lines of resolution.

In the first test, reference measurements were very similar and are not plotted. However, spectra calculated from test measurements are displayed in Figure 11. Energy within each spectrum was not evenly distributed over all frequencies, but was dispersed into groups within various frequency ranges. To compare the results of each person's measurements, the spectra were divided into major frequency ranges (or groups) which contained energy significant for evaluation. These groups are indicated by the vertical lines in Figure 11. Within each frequency range, an overall RMS acceleration value in g was obtained.

The largest spread in magnitude difference for any frequency group between 0 and 990 Hz was about 0.3 g. This value was for the range of frequencies between 885 and 990 Hz. Above 990 Hz, hand held probe measurements approached the reso-

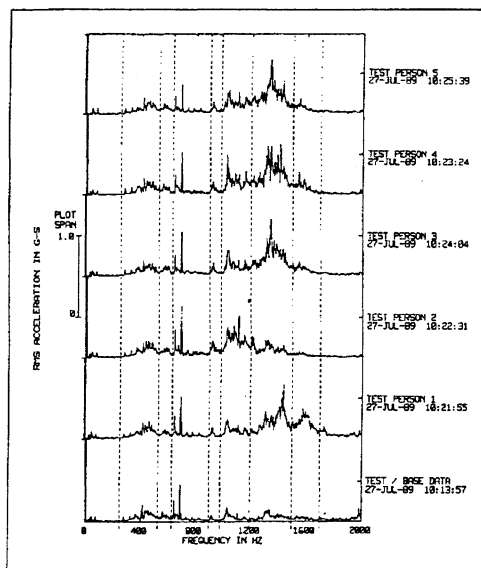


Figure 11. Test spectra comparing the measurements taken by five people using a CSI 310 hand held probe with a pointed 2 in. steel stinger attachment.

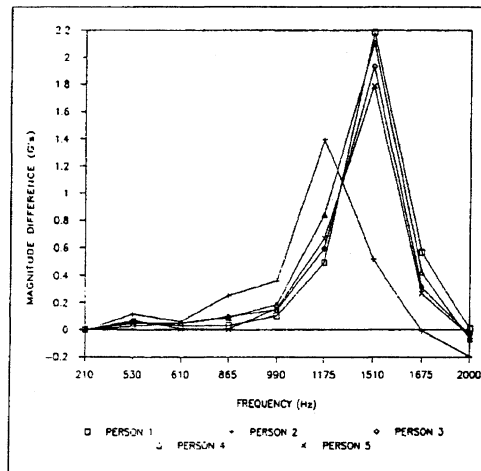


Figure 12. Deviation between the base and test magnitude difference curves derived from measurements taken with a CSI 310 hand held probe by five different people.

nant frequency of the sensor system. This phenomenon was characterized by large deviations between the reference accelerometer readings and individual readings.

The previously reported results revealed the amount of variability (or non-repeatability) between readings by each individual. To determine the accuracy of the test sensor (hand held probe), the magnitude differences calculated for each measurement within a particular frequency group were subtracted from the magnitude differences calculated for the base data of the same frequency range. The base data were considered to be the "actual" or "true" vibration measurements. A graph showing the results of these calculations is seen in Figure 12.

The hand held probe resonant frequency changed with each individual's hand pressure. One measurement (person 2, Figure 12) was substantially different from the others. The resonant frequency for this reading was approximately 1175 Hz. Other resonant frequencies, for the same hand held probe, were in the neighborhood of 1510 Hz. These results demonstrate a problem that can be encountered when taking data

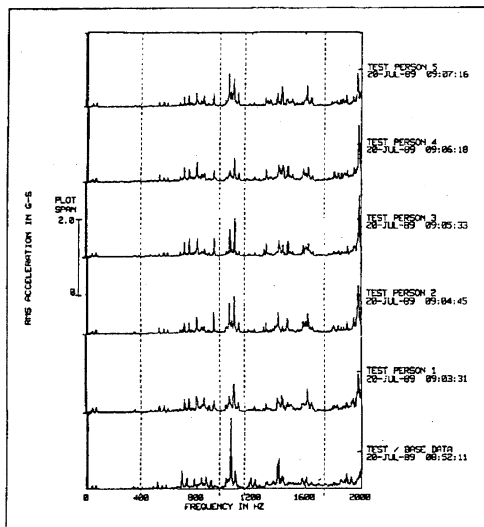


Figure 13. Test spectra comparing measurements between five people, each using the same Wilcoxon 726T accelerometer mounted on a rare earth magnet.

with a hand held probe, i.e. applied pressure to the probe will vary. The more it varies, the less repeatable and reliable the readings. Varying pressures applied to hand held probes are likely when different people take the same measurements. Even though the outlying point between 885 and 990 Hz (Figure 12) was present, the magnitude was only 0.4 g. There-

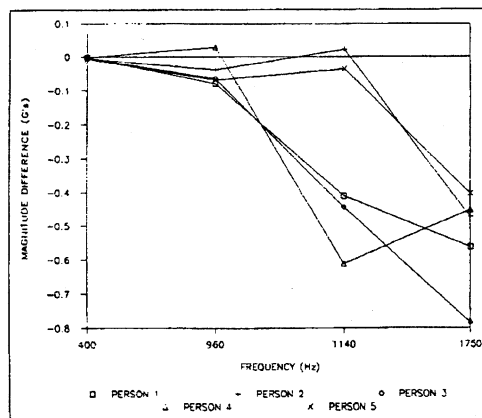
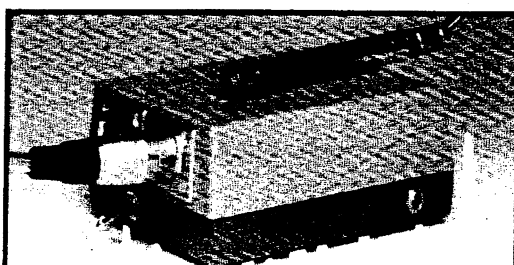


Figure 14. Deviation between the base and test magnitude difference curves derived from measurements taken by five different people with a Wilcoxon 726T accelerometer mounted on a rare earth magnet.

fore, measurements taken by various individuals are accurate for frequencies up to 1000 Hz when using a hand held probe, as long as they are taken from the same measurement position.

The second experiment included a test accelerometer (Wilcoxon 726T) mounted on a rare earth magnet. Spectra calculated from the test accelerometer measurements are plotted in Figure 13. Frequency ranges were evaluated as shown by the vertical boundary lines present in Figure 13.

Variations between the two major peaks around 1000 and 1100 Hz were observed in Figure 13. Smaller variations were also present at the frequencies ranging from 1300 to 1750 Hz.



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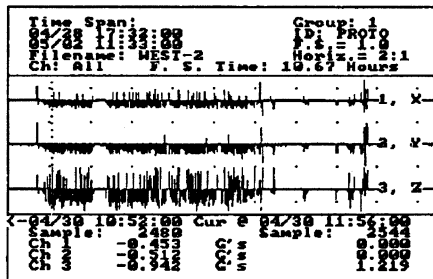
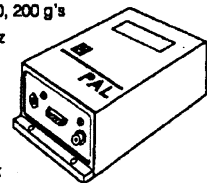
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Since the reference and test accelerometers were mounted in the same configuration, most variations are due to differences in individual placement of the magnet on the marked spot, the offset location of the test accelerometer from the reference accelerometer and/or the inducement of rocking because of improper placement.

The largest magnitude difference for any individual measurement was 0.7 g which occurred between 1140 and 1750 Hz. With respect to accuracy, the frequency range between 1140 and 1750 Hz had the greatest deviation of 0.8 g (see Figure 14). This amount was not large enough to warrant limitation of the frequency range because 0.8 g is a relatively small change over the entire 610 Hz frequency range (1140-1750 Hz). Therefore, in "real world" applications, measurements taken by different people with a magnetically mounted accelerometer can be expected to vary 0.1 g for high energy frequency intervals less than 1000 Hz. Magnitude variation within high energy frequency segments may be as much as 0.6 g for measurements up to 2000 Hz.

Variability Resulting from Multiple Measurements by One Person. In industry, one person will often be responsible for collecting data. After hours of such measurements, fatigue can be transferred into carelessness which can then affect data reliability. Therefore, a test was administered to determine how repeatable and accurate a single person can measure vibrations over a period of time. Variability in repeated readings collected by the same individual was based on 20 measurements with a hand held probe (CSI 310) at a marked position on the bearing housing (see Figure 8b). Between each reading, the hand held probe was removed. For each measurement, a spectrum using six averages and 800 lines of resolution was acquired with a frequency range of 0 to 4000 Hz.

Spectra from the test measurements are plotted in Figure 15 for the base and first seven tests. The remaining 13 replications produce spectra with similar results and are not plotted.

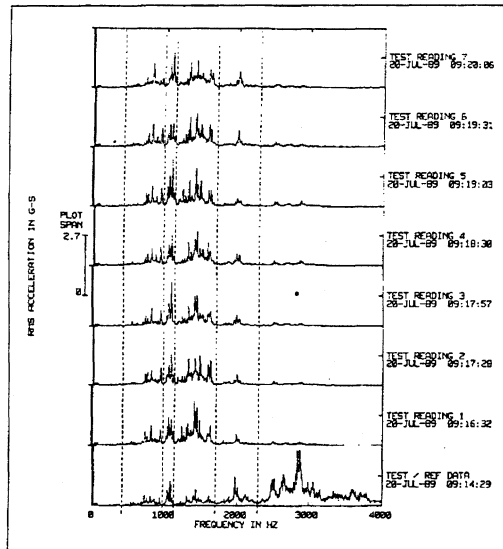


Figure 15. Test spectra of the first seven measurement replications taken by an individual using a CSI 310 hand held probe with a pointed 2 in. steel stinger attachment.

Data above 2300 Hz were not considered because very little energy was observed in this frequency region.

The largest spread in magnitude difference below 960 Hz was 0.3 g. As the resonant frequency of the hand held probe was approached, the measurements were not as accurate nor repeatable. However, when using a hand held probe (CSI 310) and measuring vibrations from a given point, data are repeat-





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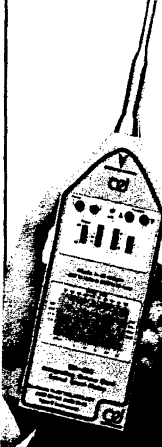
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able and accurate from 0 to 960 Hz. Therefore, the results suggest that a person can collect valid data between 0 and 1000 Hz at scheduled intervals provided machines are properly marked for repetitive data acquisition. Furthermore, when collecting field data with a hand held probe, one can expect a measurement variation of 0.3 g for high energy frequency groups in the range of 0 to 1000 Hz. Employing more than one person to collect data can increase the variability.

Recommendations

Throughout this study, many mount configurations have been evaluated. Variations which can affect the response of sensors used in these configurations were investigated as well. The main theme derived from the results of these tests is to be consistent and methodical when acquiring data for spectral analysis. Below is a list of recommendations which will enhance the chances of obtaining accurate results. Recommendations are:

General

- ☐ Limit the measurement frequency range to the linear region of the mounted sensor system frequency response.
- ☐ Limit the number of different individuals recording data.
- ☐ Allow only one person to collect data for any given route.
- ☐ Always use the same sensor and mount configuration for a particular measurement point.
- ☐ Make measurements on clean, smooth and flat surfaces.
- ☐ Take measurements in horizontal, vertical and/or axial directions.
- ☐ Mark measurement points on machinery to eliminate variation due to measurement location.
- ☐ If possible, mount a steel disk on the machine at the measurement point.
- ☐ Tighten sensors (preferably to 15 in. lbs torque) to the mount configuration to avoid looseness. This rule applies to

any configuration whether it includes a steel disk, a magnet, a quick lock or hand held probe.

- ☐ The more random the speed and load of a machine, the more averages required per measurement. For normal situations, 4 to 6 averages are sufficient. For vibrations that are more random, as many as 12 averages may be required.
- ☐ Avoid measurements where rocking may be induced. Adjust sensor position to eliminate rocking.

Magnet Mounting

- ☐ Keep magnet bottom (feet) clean to avoid rocking and insufficient contact with the machine surface.
- ☐ Do not mount hand held probes on magnets because of possible cantilever action.
- ☐ Use as light a weight of cable as feasible. The heavier and stiffer the cable, the greater likelihood of cantilever action.

Hand Held Probe Use

- ☐ Keep a constant hand pressure for every measurement taken.
- ☐ The shorter the stinger attachment, for a particular vibrating structure, the better the frequency response.
- ☐ Do not allow a hand held probe to bounce while collecting data. Bouncing will be observed as chattering in the resulting spectra.
- ☐ Avoid tilting the probe. Take measurements so that the length of the probe is perpendicular to the surface of the machine being analyzed.

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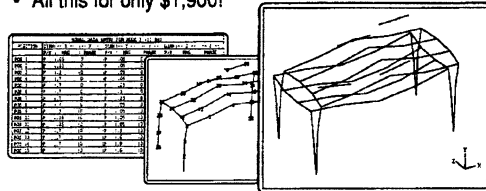
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